Deployment Guide

Planning, Installing, and Configuring the Dataupia Satori Server Solution

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May 2009

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This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to FCC 47 CFR Part 15.

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This is to certify that the Dataupia Satori Server is shielded against the generation of radio interference in accordance with the application of Council Directive 89/336/EEC, Article 4a. Conformity is declared by the application of EN 55 022:1998/A1:2000/A2:2003 Class A.

Other Emissions Conformance

Canada – ICES-003 Issue 4 Class A Digital Apparatus

Japan – VCCI Class A ITE

Australia – AS/NZS CISPR 22:2002 Class A ITE

Product and Publication Details

Model Number:	
Publication Date:	May 2009 Revision R1
Product Name:	Dataupia Satori Server
Language:	English
Publication Part Number:	DT-2008-NGDA-001

Distributed by

Dataupia Corporation

One Alewife Center

Cambridge, MA 02140

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Phone: 866-259-5971
Fax: 617-441-7776
Email: support@dataupia.com
Customer Portal: www.dataupia.com Click the **Customer Login** link at the top right of the page.

Dataupia Documentation

The Dataupia document set outlines the procedures required to integrate the Dataupia Satori Server with an existing DBMS:

Dataupia Satori Server Release Overview

- Supported Platforms
- Release Features
- Known Issues
- Fixed Issues
- Supported SQL

Dataupia Satori Server Installation and Configuration Guide - Oracle

Dataupia Satori Server User Guide – Oracle

Dataupia Satori Server Installation and Configuration Guide – SQL Server

Dataupia Satori Server User Guide – SQL Server

Dataupia Satori Server Upgrade Guide for Release 2.2

Dataupia Dynamic Aggregation User Guide

Dataupia Satori Server Error Messages Manual

Related Documentation

APC NetShelter SX 24U and 42U Enclosure

http://www.apcmedia.com/salestools/JKUR-7APSBK_R1_EN.pdf

http://www.apc.com/resource/include/techspec_index.cfm?base_sku=AR3104

http://www.apc.com/resource/include/techspec_index.cfm?base_sku=AR3100

APC Switched Rack Power Distribution Units

http://www.apcc.com/products/family/index.cfm?id=70

http://www.apc.com/resource/include/techspec_index.cfm?base_sku=AP7900

http://www.apc.com/resource/include/techspec_index.cfm?base_sku=AP7930

Safety Notice

Risk of serious personal injury or death may exist when working with or near the equipment described in this manual. Risks may include (but are not limited to):

- Risk of heavy machinery falling or tipping over due to improper handling of equipment.
- Risk of collapse of floor or supporting surface due to improper installation.
- Risk of electrocution due to exposure to high voltage or improper grounding of equipment.

  Ensure that a reliable grounding path from the rack system is maintained. This unit is intended to be connected to Earth ground. Do not defeat the integral grounding in the electrical system.
- Risk of fire due to improper or faulty wiring or protection devices.
- Do not open equipment or remove protective panels, there are no user serviceable parts inside. Refer unit servicing to the manufacturer.
- There may be other risks that cannot be reasonably foreseen.

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About This Document

This document describes how to plan for, install, and configure the Dataupia Satori Server Solution. It includes a general description as well as a description of each Solution component, as well as networking diagrams. It is intended to assist customers by explaining the stages and processes involved in the physical setup and configuration of the blades and arrays of the Dataupia™ Satori Server.

Dataupia Satori Server Overview

The Dataupia™ Satori Server increases the data storage capacity, performance, and scalability of your existing Oracle or Microsoft SQL Server database management system (DBMS), enabling it to work effectively with the very large data sets required in today's business environment. Satori Server smoothly integrates with each DBMS to extend its capabilities through massively parallel processing and high-capacity, highly scalable storage.

Satori Server's Omniversal Transparency™ (OT) enables these gains without requiring changes to or disrupting the operation of your existing application or DBMS. Transparency is achieved at several layers within the data warehouse architecture:

- The DBMS sees the Satori Server as a native extension of its own table space.
- Data center operational personnel see it simply as another database server.
- The application's end user need not be aware of the Satori Server at all, as its use as a data source by the DBMS is completely transparent at this level.

Database operations are shared seamlessly across the combined capabilities of the DBMS and the Dataupia Satori Server, optimizing query execution for the highest possible performance. Increasing the storage capacity of Satori Server automatically increases its processing power, and its record storage structure is easily extended and reconfigured, so you can scale with minimal effort as you add to your data sets and transfer data from existing storage array infrastructures.

Each Satori Server consists of named logical array of blades that provide storage pooling and a single point of management. The ability to link multiple database instances on your DBMS to each array, and to change these links as required, lets you make the best use of your resources in addressing current needs and rapidly adjusting to changing business requirements.

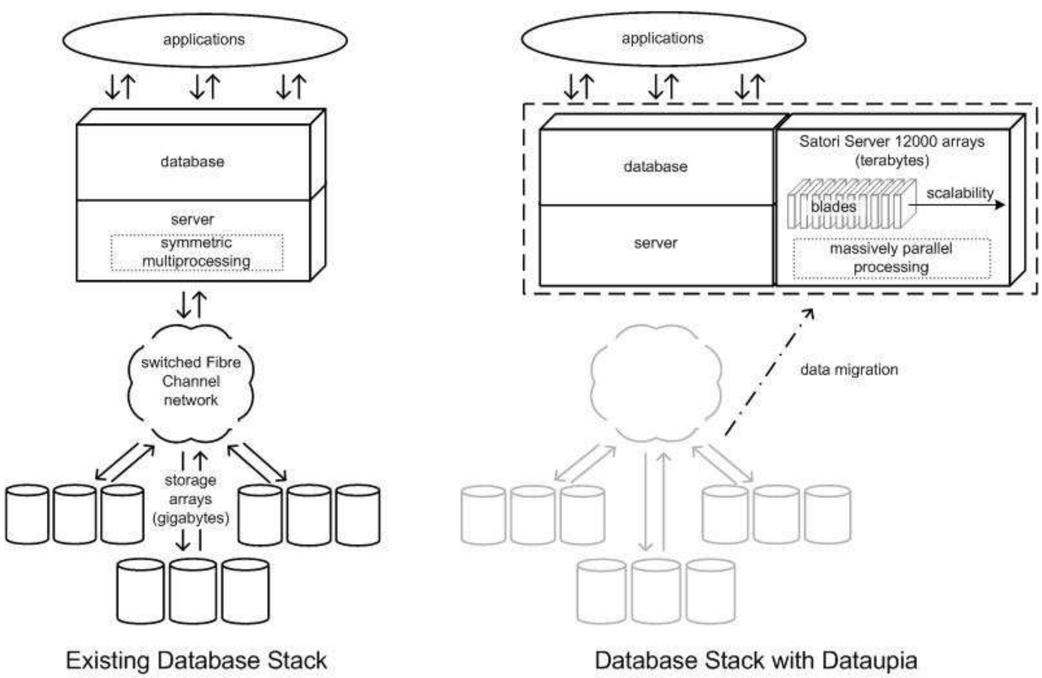

Figure 1 – Data Migration to Satori Server

Figure 1 shows how the Dataupia Satori Server extends the capabilities of a database installation without requiring any modification of the existing application/DBMS.

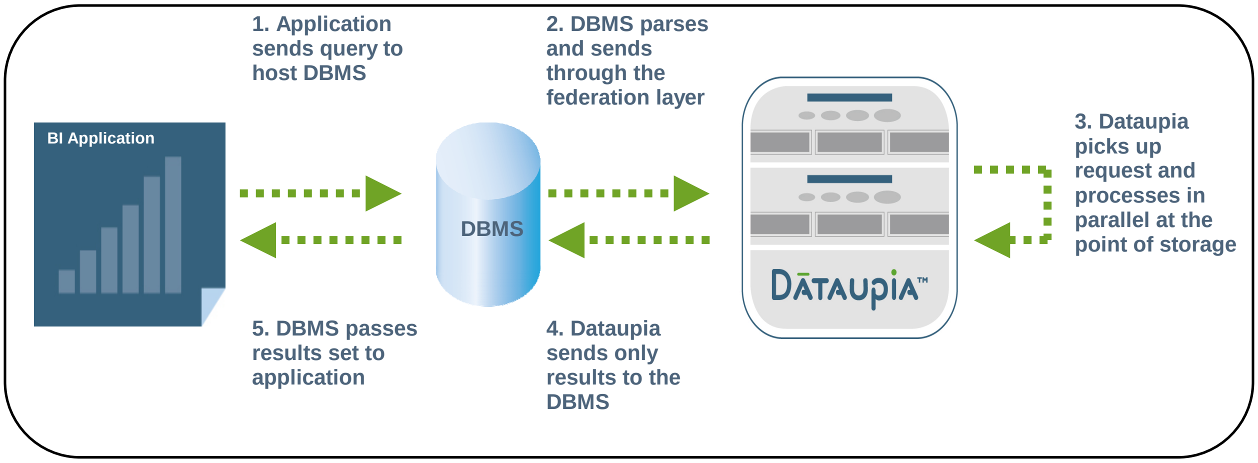

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Figure 2 – Satori OMNIVERSAL TRANSPARENCY™

Solution Components

- **Dataupia Satori Server**

  The Dataupia data management system is an all-in-one solution – server, storage, and optimization software packaged as a single appliance – and the first and only solution specifically designed with RDBMS and application transparency. Dataupia delivers persistent access to as much data as an organization needs. The combination of highly specialized software and powerful processors allows large amounts of data to remain online and ready to be accessed by reporting or analytics applications. The appliance easily dovetails with your existing data center infrastructure to quickly augment the reach and performance of applications that depend on data. All of the blades in a Satori Server must be interconnected by an Ethernet network, but the only other servers which need to communicate with them are the host DBMS (Oracle or MS SQL Server), the Loader server(s), DA server(s), or other optional server(s) that have Dataupia client software installed.

Figure 3 -- A Satori Server Blade

- **Loader Server (Optional)**

  The Dataupia Data Loader loads data from files into tables on a Dataupia array. Dataupia can provide the loader as software or as an extra server in a that fits into the rack. The Data Loader uses the same hardware as the Satori Server, but is configured to optimize the loader utility.

- **Dynamic Aggregation (DA) Server (Optional)**

  Dynamic Aggregation is available as an option for the Dataupia Satori Server. The combination of the Dataupia Satori Server with Dynamic Aggregation is geared to delivering high performance for routine BI and reporting tasks against massive amounts of data. It rapidly assembles and refreshes multidimensional views of large data sets and makes them available to thousands of concurrent users.

- **Aggregation Server (Agg server)**

  Not used with software release 2.1 and later.

  A Satori Server that performs aggregation functions on final query results. With software release 2.1 and later, the functions performed by the Agg server are shared by all Satori Servers in the array.

- **Network switch**

  A 48-port managed network switch configured with VLANs that is used to provide connectivity for all servers and components in the solution.

- **Combrio remote support (optional)**

  Combrio is a remote support solution which provides an alternative inbound VPN for remote access to the Satori Server components for Dataupia support personnel.

- **Terminal server (optional)**

  A RS-232/422/485 serial-to-Ethernet connection that provides console access to each blade as a backup or alternative method.

- **Keyboard, Video, Mouse switch (KVM)**

  A rack-mount console to provide onsite access

- **Switched Rack Power Distribution Units (PDUs)**

  Switched rack Power Distribution Units that provide load metering combined with controlled on/off switching by individual outlets for remote power recycling, delayed power sequencing of equipment and outlet use management.

Blade Specifications

These specifications are for a single blade server. This information applies to the Satori Server, the Dynamic Aggregation Server, and the Loader Server. A complete Satori system will consist of more than one blade. Specifications are subject to change without notice.

Form factor	2U chassis suitable for mounting in industry standard EIA-310 (19 inch rack). Mounting rails and hardware are included with the server. Height 3.5 in./ 8.9 cm x Width 17.5 in. / 44.45 cm x Depth 27.3 in. / 69.4 cm
Disk storage	Eight hot-swappable drives (accessible from the front panel), 500GB Enterprise SATA or equivalent.
Satori Server	1.7TB; 7 drive RAID5 + hot spare
DA Server	2.5TB; 2 drive RAID1 + 6 drive RAID5
Loader Server	1.5TB; 2 drive RAID1 + 6 drive RAID10
CPU	2 x AMD Dual core High Efficiency (HE) Opteron processors
Memory	8 x 1 GB DDR2-667 Registered ECC
OS	64 bit
Power	115-240 VAC single phase, 47-63 Hz, 300 watts operating, 510 watts max; IEC 115V/10A cord set with ground for US applications. This is for a single Satori Server. Consult an electrician to assist in determining the circuit(s) required to meet local electrical codes.
Operating temperature	41º F to 95º F / 5º C to 35 º C maximum 50º F to 77º F / 10º C to 25º C recommended
Humidity range	10-80% (non-condensing)
Cooling	Data center air conditioning and climate control. No additional ventilation or cooling is required.
Maximum altitude	6,000 ft. / 1,829 m
Weight	47 lbs / 21.4 kg (chassis) 63 lbs / 28.6 kg (shipping)
Certifications	CSA, FCC, CE, RoHS Compliant

Table 1 - Satori Server specifications

Pre-Racked Satori Systems

Dataupia provides two racked configurations of solutions components providing for up to 16 blades per rack. Multiple racks may be joined to achieve larger configurations. Each rack ships assembled and wired for the blades and includes power distribution units, a network switch, a terminal server and a KVM. Final assembly at the site involves unpacking the blades, sliding them into the rack (the rails are pre-installed on the blades and in the rack), and connecting the cables. This approach significantly reduces installation time, allows for future upgrades, and provides a known and supportable hardware combination. See Table 1 for Satori Server specifications. See Table 2 for rack specifications. See Tables 3 and 4 for 24U and 42U rack configurations.

The Loader Server and Dynamic Aggregation Server are optional components that may be part of the solution designed for your enterprise data requirements. Each of these servers has the same hardware specification as the Satori Server and occupies one server position in the rack.

Customer-owned equipment can be added to the rack at unused locations as long as available power and weight restrictions are not exceeded. Pre-racked configurations may be customized for OEM applications.

Figure 4 – Pre-Racked Satori System with Omniversal ™ Transparency

Satori Rack Specifications[1] (120VAC USA models)

	24U rack	**42U rack**
Number of Servers. Servers can be any combination of Satori, DA and/or Loader servers	1-8	1-16
Power Connector	2 x NEMA 5-15P	2 x NEMA L5-20P
Power cord length (from PDU)	12 ft (3.66 m)	10 ft (3.05m)
Network Switch	48 port 1Gbps	48 port 1Gbps
Terminal Server	Digi Portserver TS16	Digi Portserver TS16
KVM	Standard	Standard
Combrio Remote access	Optional	Optional
Number of Power distribution units (PDU)	2	2
Net Weight (0 servers/8servers)	331 lbs/707 lbs	410 lbs/1,162 lbs
Dimensions (inches/mm)	47.19H x 23.62W x 42.13D 1198H x 600W x 1070 D	78.39H x 23.62W x 42.13D 1991H x 600W x 1070 D
Net depth with stabilizing feet	N/A	50.3 inches 1278 mm
Shipping Weight (0 servers installed) (Note: 24U racks ship with blades installed)	370 lbs	453 lbs
Shipping Dimensions (inches/mm)	53.15H x 35.39W x 47.99D 1350H x 899W x 1219	83.40H x 35.40W x 48.00D 2118H x 899W x 1219
Weight Capacity Static/Dynamic	3000 lbs / 2250 lbs	3000 lbs / 2250 lbs
Minimum mounting depth	7.52 inches 191 mm	7.52 inches 191 mm
Maximum mounting depth	36.02 inches 915 mm	36.02 inches 915 mm
Rack: APC NetShelter SX model (black with sides and locking doors and sides)	24U	42U
Conformance	EIA-310-D, UL 60950, RoHS	EIA-310-D, UL 60950, RoHS
PDU: APC Switched Rack PDU model no	AP7900	AP7930

Table 2 - Pre-Racked Satori Specifications (less servers)

[1] Specifications are subject to change without notice.

1U Position	Equipment
42	Ethernet Switch #1
41	blank
40	Terminal Server
39	Combrio remote support (optional)
36 - 38	blank
24 - 35	Servers 11-16
23	blank
22	KVM
21	Unused
1 – 20	Servers 1-10

Table 3 - 42U pre-racked configuration

Position	Equipment
24	Ethernet Switch
23	Terminal Server
22	KVM
19 - 21	blank
3 – 18	Servers 1-8
2	PDU2
1	PDU 1

Table 4 - 24U pre-racked configuration

Customer Supplied Equipment Racks

Dataupia uses APC NetShelter SX rack systems, but the system components will fit into a standard 19” wide rack. If you prefer to supply your own rack(s), you may need to adapt the supplied mounting hardware or cables, or obtain mounting hardware or cables from another source.

Environment

Data center air conditioning and environment control is required for the Satori Server system.

- The maximum recommended ambient temperature is 25° C. The internal temperature of each rack should be considered for continued safe operation.
- The minimum recommended ambient temperature is 10° Centigrade.
- Unimpeded airflow is required to the front and rear of the unit (server). Do not block or restrict airflow when installing a unit in a rack.

Planning for Deployment

This section discusses the architecture and major concepts involved in planning Satori deployments.

Satori Client

From Satori's point of view, a client is a host which has Satori client software installed on it. Satori client software enables connection to the Satori Server array, and includes Satori Omniversal Transparency ™ (OT) for database hosts and Satori utilities. It can be installed on host database servers and loader servers. Enterprise users will continue to use their existing host DBMS "clients" and applications as before, and may be unaware that the underlying relational database is using Satori OT to run their queries on a Satori array.

Satori Client software is available for:

- Solaris SPARC – enables OT Oracle 10g/11g* and/or Satori Utilities.
- Linux 64-bit – enables OT Oracle 10g/11g* and/or Satori Utilities.
- Windows Server – enables OT SQL Server.

Connection to the Satori Server is established using Satori client software. No other drivers or configuration steps are required. Satori implements proprietary IP communication protocols that perform discovery and communicate with other Satori devices. These methods do not utilize classical network discovery mechanisms such as DNS. Instead, individual Satori Servers identify themselves to each other and to client software using a server registration ID and group membership tokens. Satori Server discovery is possible only from those hosts that are directly connected to the same subnet as the Satori Servers.

Client software also installs Satori utilities (except for on the Windows platform). Client software can be installed on a machine to enable host database OT, to enable utilities, or both. Satori clients which are critical components – such as host database servers – will have a dedicated NIC which attaches to the Satori "DATA" subnet.

* check for availability

Host Database Server(s) and Satori Omniversal™ Transparency connector

Database servers that need to communicate with one or more Satori Servers must be Satori clients. There is no limit to the number of Satori Servers (arrays) that can be accessed by an Oracle instance. Each Oracle to Satori "link" requires configuration of an instance of the Satori client Oracle Connector. Refer to the client installation instructions for installation details.

For Oracle, Satori client installation involves modifying the listener.ora and tnsnames.ora and a restart of the Oracle listener. The Satori client installation program performs these modifications automatically based on user input. If possible, schedule the Oracle configuration steps of the client installation during a maintenance interval to avoid disrupting the database.

For Oracle, the Satori client using Omniversal ™ Transparency (Oracle Connector) requires the Oracle Java VM.

Installation of the Satori client requires **root** or **sudo** and DBA access to the database.

Dataupia recommends the use of a dedicated 1Gbps NIC (minimum) to connect the database server(s) to the Satori array backbone network.

Satori Utilities

dtldr, dtlscan – Data Loader is able to load a wide variety of data formats into Satori tables

dtunload – unload data from Satori to a csv file

dttable – table creation, modification, destruction utility

regtable, unregtable, chtable – data distribution control utilities dtsql – sql client

mkdistmap, gendistmap, lsdistmap - distribution

lssystem, lsblade, lsslice, lstable, lsservice – query server properties and status

Dataupia Management Console (DMC)

Dataupia Management Console (DMC) is a web-based client that provides monitoring and control functions for Satori Server blades, and can be accessed using a Web browser. A user name and password are required to gain access. DMC access is generally exposed to the customer's internal network, and requires an IP address to be assigned. See Chapter 3 of the *Dataupia Satori Server User Guide* for information on how to use the DMC.

Satori Server Networking 101

Figure 5 depicts the architecture of a four node Satori Server (the grey inset in the figure) as installed in an enterprise network with existing host DB servers. Two existing Oracle hosts have been attached to the Satori DATA network with the installation of Satori client software and a second NIC, enabling the Oraclee hosts to pass SQL to the Satori using OT™. Also a new dual-homed server was added to run the Satori Loader.

Satori has added a backend 'DATA' network which interconnects the servers, and also connects all hosts which have Satori clients. This backend network can either be physically separate from the customer network, or it can be the same network. There are pros and cons with either approach:

- Physically separate networks:
    - Pros: best performance, isolation, and backend addressing and traffic doesn't impact customer networks (and vice-versa).
    - Cons: more complex network design.
- Shared networks:
    - Pros: simplest network design.
    - Cons: least isolation.

Dataupia's recommended best practice is to establish physically separate networks. A network switch is provided in the pre-racked Satori solution for this purpose, or you can provide your own switches.

Three untagged V-LANs are defined on the switch. They are:

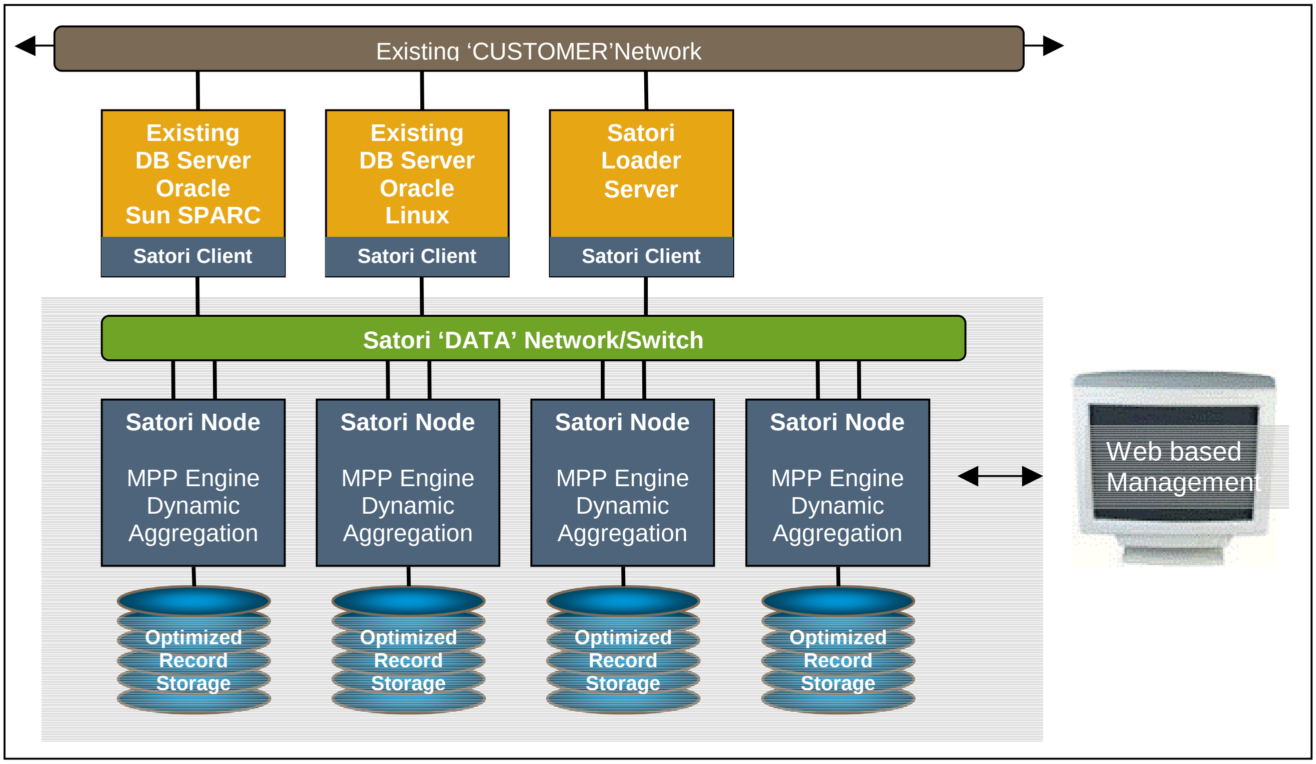
- 'CUSTOMER' VLAN—used for connections to the customer's network.
- 'DATA' VLAN—the Satori backbone used for queries and data
- 'HEARTBEAT' VLAN (optional, not shown in the diagram)—used for inter-server communication of maintenance and control messages on larger arrays. In installations where the HEARTBEAT network is not needed, this traffic transparently uses the DATA network. In installations which employ it, this network is "zero-conf" by default (169.254.x.x/16) and therefore does not require configuration or the reservation or assignment of IP addresses.

These three VLANs are separate (logical) networks. It does not matter if they are implemented as VLANs or if they are on discrete switches. Dataupia recommends the use of enterprise-grade 1Gbit switch gear and CAT-6 cabling for these networks. HEARTBEAT may use 100Mbit class equipment.

Figure 5 depicts a typical interconnect schematic diagram of the Satori server and its peripherals. It shows how each of the different type of Satori servers (Satori, DA and Loader and Aggregation servers) are typically connected to the networks. It's important to remember to multiply by the number of servers of each type in your proposed solution configuration when calculating the total connection requirements.

Customers can establish IP routing from the CUSTOMER network to the DATA network if they would like to use another host or firewall, however, it is important that traffic passing from the DB Server, Loader server and DA server to and from the Satori Server array NOT pass thru a firewall, as this may have a negative impact on query performance. But if routing is established, non-performance-critical Satori clients can be connected to routable networks within the customer's enterprise.

Use the worksheet found in Appendix I



[A3]

Figure 5 - Satori Architecture

IP Address requirements

In order to install Satori Server, IP addresses will be needed for the following components. Depending on the final network design, the addresses may fall under either the customer network or the Satori network. Table 5 below shows the recommended requirements that are typically used with pre-racked solutions and a closed Satori DATA network segment.

	Satori network	CUSTOMER network
Satori Servers	1 IP address for each Satori server + 1 management address	none
Satori client machines	1 IP address for each dedicated NIC	1 each dedicated NIC
Network switch mgmt address		1
Terminal server remote access		1
PDU control		2
Combrio remote access	TBD	TBD
DMC provisioning	0	1

Table 5 – Typical system IP requirements for pre-racked solutions

	Eth0	Eth1	Eth2	Term	Com
Satori Server	CUST[2]	DATA	HB	KVM	TS
Loader Server	N/C	DATA	CUST	KVM	TS
Dynamic Agg Server	N/C	DATA	CUST	KVM	TS
Aggregation Server[3]	N/C	DATA	HB	KVM	TS

Legend:
CUST – CUSTOMER network
DATA – Satori DATA network
HB – Satori HEARTBEAT network
KVM – keyboard/video monitor
TS – Terminal Server
Notes:
Eth0 is 10/100 Mbps Ethernet
Eth1 & Eth2 are 1Gbps Ethernet
Term is the local VGA port
Com is the local communications (RS232) port. The Satori Server OS console is accessible on this port at all times.

Table 6 - Satori components connect requirements

Sample IP configuration

Assume the customer network is existing and is setup using:

Network address	10.214.128.0
Netmask	255.255.252.0 (22 bit mask)
Default gateway	10.214.128.1
DNS	10.214.128.2, 10.214.128.3
NTP	10.214.128.4

The Satori DATA network can be setup using:

Network address	10.0.0.0
Network mask	255.255.0.0 (16 bit mask)
Default gateway	Not required
DNS	Not required
NTP	10.0.2.10 (enable on this interface)

Component	Satori Network	Customer Network
Host DBMS Server	`10.0.2.[10-19]`	`10.214.128.10*`
Loader Servers	`10.0.2.[20-29]`	`10.214.128.200`

2 Only one blade in a Satori server array needs a connection from eth0 to the CUSTOMER network. This is for DMC provisioning to the customer's network.

3 The Aggregation server is no longer used in software release 2.1 and later.

DA Servers	`10.0.2.[30-39]`	`10.214.128.201`
Satori Servers	`10.0.1.[2-252]`	N/C
DMC mgmt	`N/R`	10.214.128.202
Satori Server mgmt IP	`10.0.1.253`	NA
Satori Switch mgmt	`N/C`	10.214.128.204
Term server	`N/C`	10.214.128.205
PDU-1	`N/C`	10.214.128.206
PDU-2	`N/C`	10.214.128.207

Table 7 – Sample IP Configuration, pre-racked solution

Email Alerts

Satori Servers report critical conditions by sending alerts via email. Alerts are configurable and can be sent to a list of internal or external recipients. Enabling these alerts is an important way to maintain the health of the server, and Dataupia encourages clients to forward email alerts to our alert facility for analysis. Each server reports its own alerts independently of the other servers, so it is possible to receive multiple alerts for conditions which affect more than one server. In order to send email alerts, Satori Server blades require the IP address and port of an SMTP server. The content of email alerts is limited to the internal indicators necessary to specify the nature of the problem – no user data is included in the alert messages. Here is a partial list of the available email alerts:

- Raid array fault
- Temperature too high or too low
- Free data space too low
- Free file system space too low
- Too much memory in use
- Paging activity too high
- Disk I/O too high
- Network utilization too high

See Appendix F for more information about email alert configuration.

Remote Maintenance

Maintenance and support personnel use SSH to gain remote access to the Satori server blade(s). In the event that a server doesn't respond to SSH, it can also be controlled via a telnet connection to a pre-assigned port on the terminal server. Remote support is generally achieved using VPN or SSH connection supported by the customer enterprise. Access to a host which is connected to the DATA subnet (described later) is necessary to maintain the Satori Server. Usually a host that is connected to both the CUSTOMER and DATA subnets – such as the Loader server – is used.

NTP

Availability of an NTP source for Satori Server is recommended.

SNMP

SNMP is not currently supported, look for it in a future Satori Server release

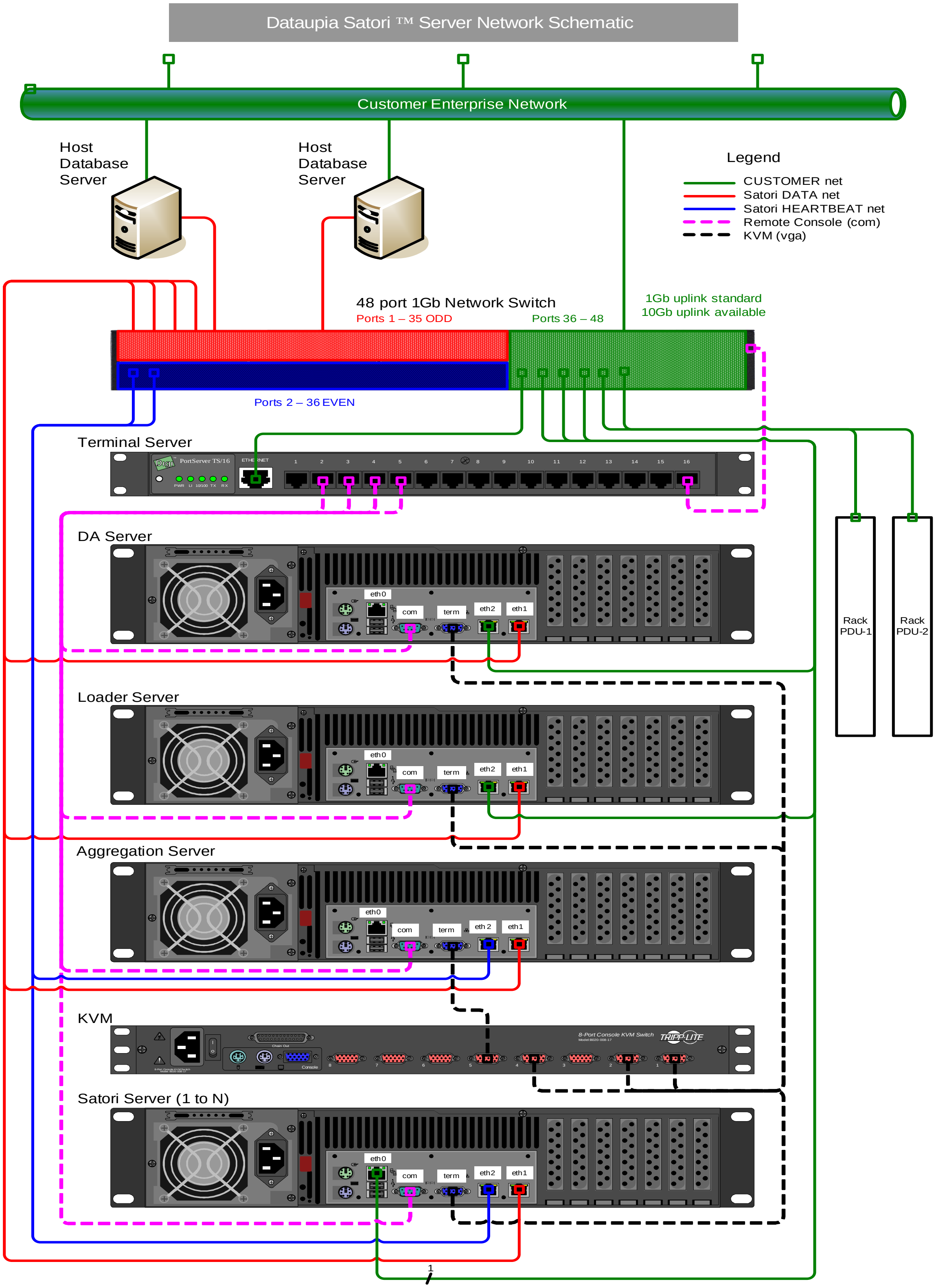


Figure 6 - Satori Server Network Schematic

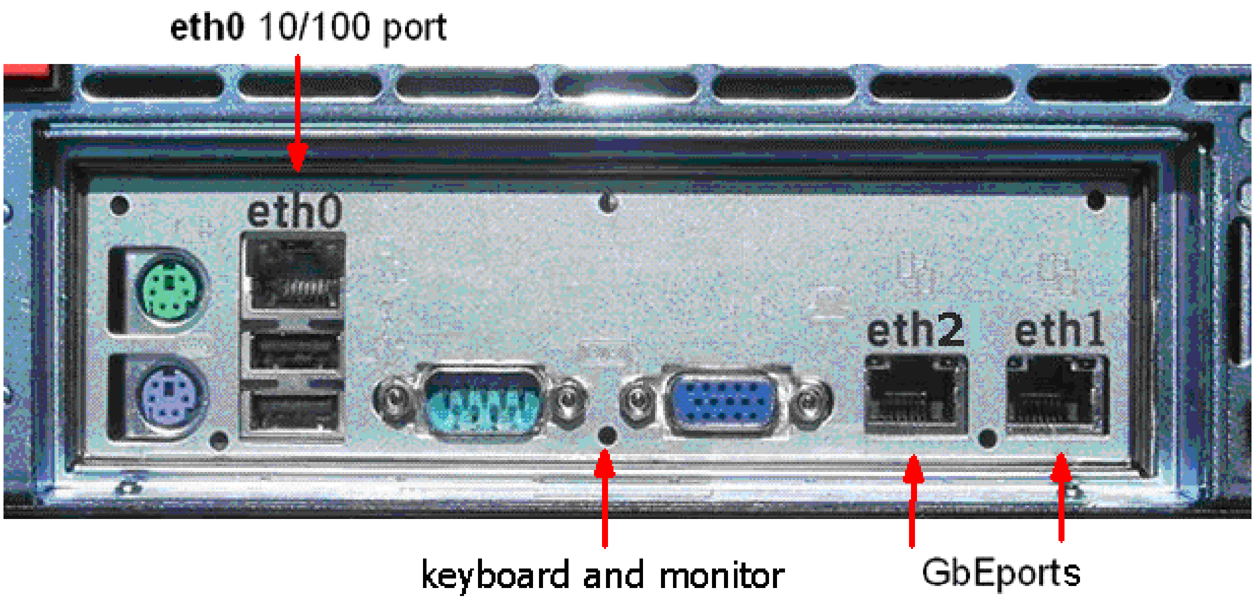


Figure 7 - Blade Connections (rear)

Deployment Checklist

This is a review of important issues regarding deployment planning for Satori:

- System Sizing: Dataupia will help you determine the right combination of Satori solution components.
- Host Databases: identify and check those which require Satori client installation
- Network design: DATA network
- Network design: CUSTOMER network, type of customer uplink requirements
- Rack layout: customer equipment in the rack?
- Physical Planning: power requirements, footprint.
- Power space
- Customer network uplink and IP: provisioning
- Host Databases: provision IP and connection(s) to Satori network.
- Host Databases: install dedicated NIC for Satori
- Host Databases: install Satori client software

Pre-Racked Satori Installation

These instructions are for setting up a pre-racked Satori system. This information only applies to Satori racks which have been pre-assembled at the factory complete with server mounting hardware, network and power cabling, and pre-racked peripheral equipment. If you have ordered a rack that is not factory pre-assembled and wired, see Appendix A for instructions on how to assemble it.

Mechanical assembly of the pre-racked Satori system is straightforward. After you remove the rack from the shipping pallet, slide the servers into the rack, install the retaining screws at the front of each server, and then connect the cables to each server. Servers are typically numbered in ascending order from the bottom of the rack. Refer to the wire run list shipped with the system.

- **Important:** Remove the rack from the shipping pallet before installing the servers. Do not attempt to do this alone; damage to the system or personal injury could result. After the system has been assembled, be sure not to pull out more than one blade at a time, as there is a risk of tipping the rack.

Required tools:

#1 Phillips screwdriver

Adjustable wrench

Parts kit supplied with the system for assembly and configuration

Assembly and setup steps for a pre-racked Satori system:

Step 1 Read the safety notices and wear protective glasses and clothing.

Step 2 Unpack and remove the rack from the shipping pallet. Do not attempt to perform this step alone; personal injury or damage to the system may result.

Step 3 Move the rack into place.

Step 4 Unpack the power cable(s) and route them as required, but do not connect to a power source at this time.

Step 5 Use a wrench to adjust the four leveling legs.

Step 6 Unpack the Satori Server blades one at a time and load them into the equipment rack starting from the bottom of the rack. Install the two retaining screws (M6 black) into the top hole at each side of each server's front bezel.

Step 7 Connect the cables to each server as specified in the supplied wire list and by the tags on the cables. Each cable (including power cables) is labeled on both ends.

1. Grey CAT-6 to ETH1
2. Blue CAT-6 to ETH2
3. KVM to D-15 (console)
4. Black CAT-6 (with D-9) adapter to D-9 to serial port.
5. Power cable (be sure to engage the strain relief clip)

Step 8 Connect the power cords to main power.

Step 9 Power up all units one at a time. Wait 30 seconds between each unit to allow time for the disk drives to spin up. Check the power load on each PDU display for proper load and balance. If necessary, adjust by redistributing the load.

Step 10 Configure the network switch; the procedure can be found here: Network Switch Configuration.

Step 11 Configure the terminal server; the procedure can be found here: Configure the Terminal Server.

Step 12 Configure Combrio remote access; the procedure can be found here: Power Up and Configure the ComBrio Box.

Step 13 Set the IP addresses on the PDUs; the procedure can be found here: Configure the APC Power Distribution Units (PDUs).

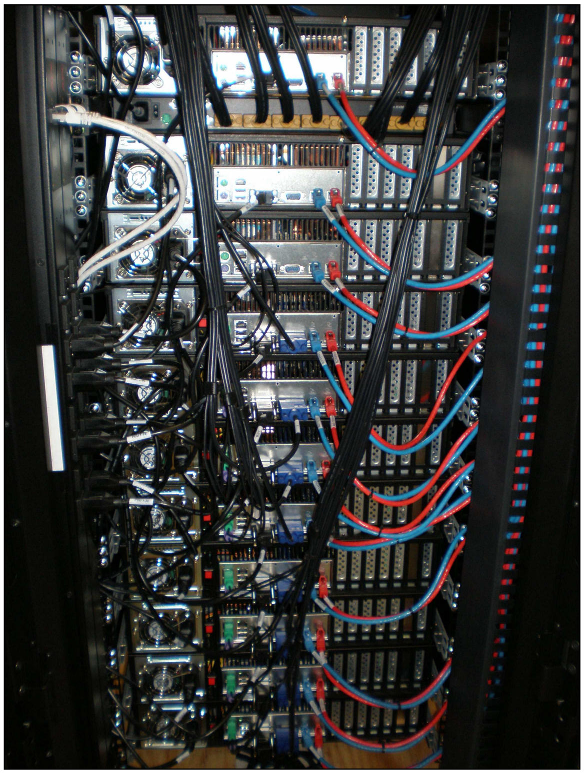
Step 14 Ask the customer DBA or network specialist to configure the Satori-facing NIC on the host database.

Step 15 Configure the Customer and Satori-facing NIC on the Loader server and DA servers; these instructions may help: Appendix C: Setting IP addresses.

Step 16 Configure the Satori Server array; the procedure can be found here: Configure the Satori Server Array.

Step 17 Configure the DBMS connector on the Host Database; the procedures can be found here: Install and Configure the Oracle Connector and here Install and Configure the Dataupia MS SQL Server Connector.

Step 18 Test connectivity and array access.



Network Switch Configuration

In this section you will install and configure the network switch for the Satori network. If you have not yet completed the design of the Satori network, do so now by referring to the section Planning for Deployment.

WARNING: Before starting this procedure, disconnect the network switch from all customer networks (including any uplinks), and consult with the customer before proceeding.

The network switch will be in one of two states, factory default or preconfigured. If the switch is preconfigured, there will be a sticker on the unit indicating its current IP address. If the switch is not preconfigured, skip to section Factory Default Switch.

Pre-configured Switch

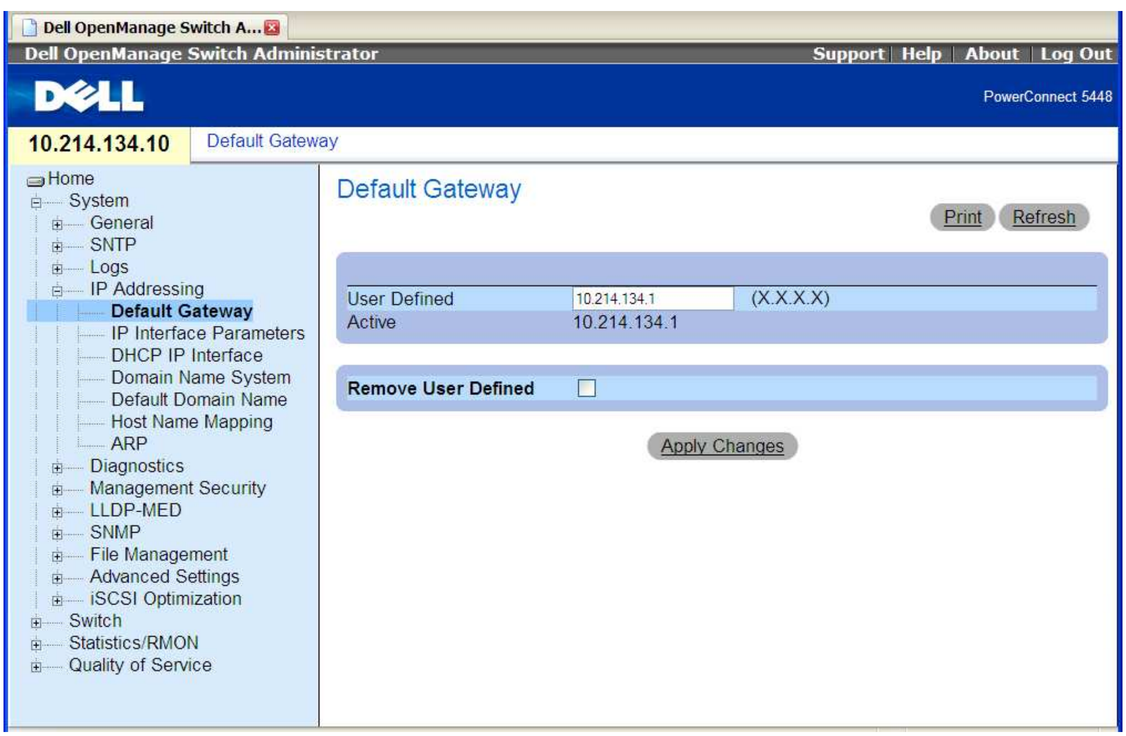
To get started, temporarily assign an IP address to your laptop that is within the range of the switch's *old* IP address (listed on a label on the switch). For example, use 10.214.134.50, 255.255.255.0. Connect an Ethernet cable from your laptop into a port in the CUSTOMER VLAN on the switch; (green section, ports 37 - 48, see Configuring VLANs for Satori). Once the IP on your laptop is set, confirm you can *see* the switch by pinging the IP address on the label.

Connect to the GUI of the switch by browsing to the switch's *old* IP address and change the IP address, subnet mask address, and default gateway of the switch to the provided customer IP addresses. Save the new settings and the configuration. The steps are as follows:

In a browser, navigate to `http://<switch_ip _address>`

Login to the switch: admin/bigdata

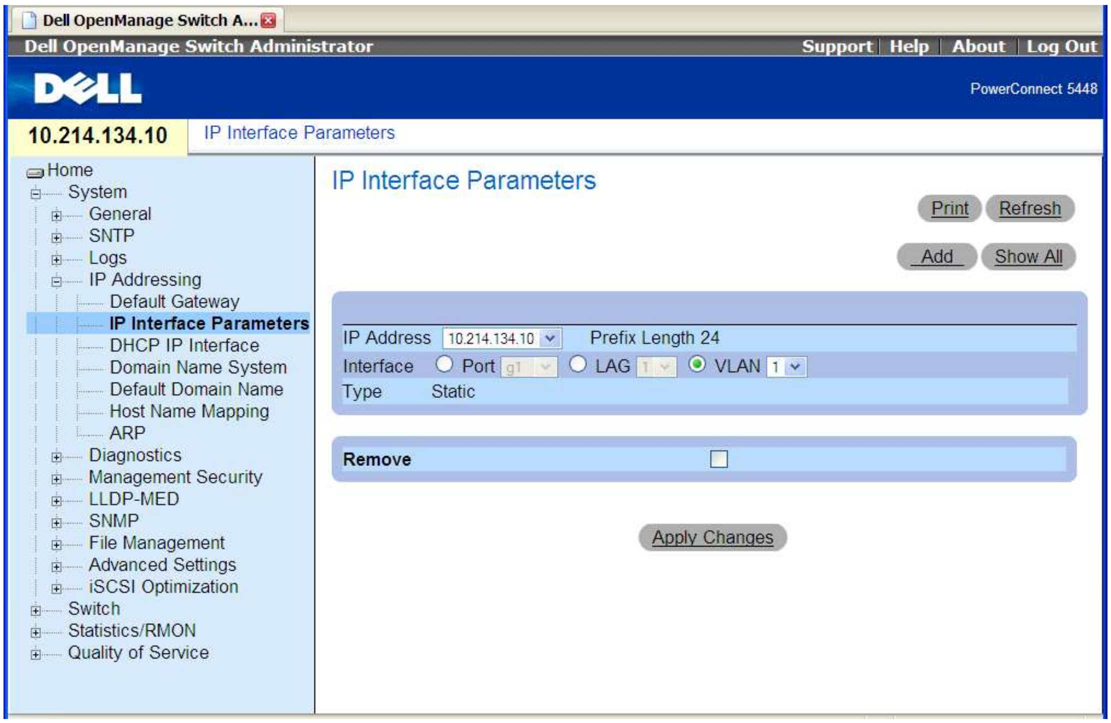
To change the default gateway, click on **System > IP Addressing > Default Gateway**

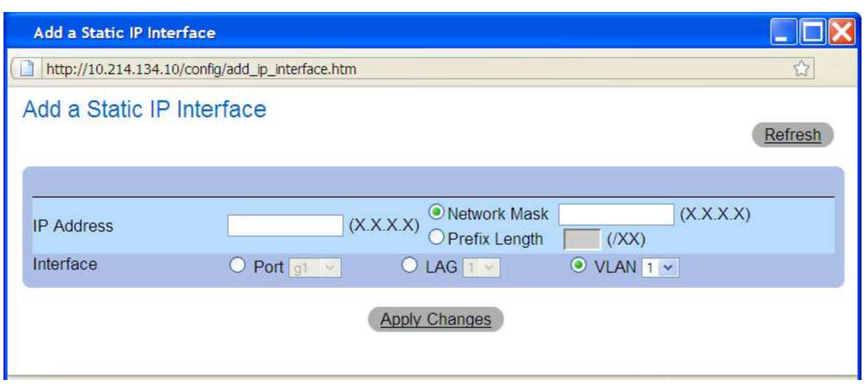


Enter the customer-provided default gateway for their network, then click **Apply Changes** to save your changes.

To change the IP and subnet mask, select: **System > IP Addressing > IP Interface Parameters**.

Next, you will see the following IP Interface Parameters screen. Click **Add** in the upper right corner to add the new customer-provided IP address for the switch and its subnet mask.





Enter the customer-provided IP address for the switch and its subnet mask. Confirm that the VLAN is selected and 1 is specified for the Interface setting. Click **Apply Changes** to save your changes.

Finally, after changing your local laptop's IP to be within the provided IP range, ping the switch's IP address and the customer default gateway address to confirm connectivity.

Factory Default Switch

If there is no indication of the switch's IP address, it is set to the factory default and will need to be configured. The steps for this are as follows:

Using the supplied 9pin-to-9pin cable, connect to the switch, then open a Console session to the switch. Plug in the switch to power it on, paying close attention to the boot process. The initial device configuration is through this Console port. After the initial configuration, the device can be managed either from the already connected Console port or remotely through an interface defined during the initial configuration.

If this is the first time the device has booted up, or if the configuration file is empty because the device has not been configured, you will be prompted to use the **Setup Wizard**. The **Setup Wizard** provides guidance through the initial device configuration, and gets the device up and running as quickly as possible. You can skip the Setup Wizard and manually configure the device through the device CLI mode.

The Setup Wizard configures the following fields.

- SNMP Community String and SNMP Management System IP address (optional)
- Username and Password
- Device IP address
- Default Gateway IP address

The following is displayed:

```
Welcome to Dell Easy Setup Wizard

The Setup Wizard guides you through the initial switch configuration, and gets you up and
running as quickly as possible. You can skip the setup wizard, and enter CLI mode to
manually configure the switch.

The system will prompt you with a default answer; by pressing enter, you accept the
default.

You must respond to the next question to run the setup wizard within 60 seconds,
otherwise the system will continue with normal operation using the default system
configuration.


Would you like to enter the Setup Wizard (you must answer this question within 60
seconds? (Y/N)[Y]Y

You can exit the Setup Wizard at any time by entering [ctrl+Z].
```

If you enter [N], the Setup Wizard exits. If there is no response within 60 seconds, the Setup Wizard automatically exits and the CLI console prompt appears.

If you enter [Y], the Setup Wizard provides interactive guidance through the initial device configuration.

Note: If there is no response within 60 seconds, and there is a BootP server on the network, an address is retrieved from the BootP server.

Note: You can exit the Setup Wizard at any time by entering [ctrl+z].

Wizard Step 1

The following is displayed:

```
The system is not setup for SNMP management by default.
To manage the switch using SNMP (required for Dell Network Manager) you can
        - Setup the initial SNMP version 2 account now.
        - Return later and setup additional SNMP v1/v3 accounts.

For more information on setting up SNMP accounts, please see the user documentation.
Would you like to setup the SNMP management interface now? (Y/N)[Y]Y
```

Enter [N] to skip to Step 2.

Enter [Y] to continue the Setup Wizard. The following is displayed:

```
To setup the SNMP management account you must specify the management system IP address
and the "community string" or password that the particular management system uses to
access the switch. The wizard automatically assigns the highest access level [Privilege
Level 15] to this account.
You can use Dell Network Manager or CLI to change this setting, and to add additional
management systems. For more information on adding management systems, see the user
documentation.
To add a management station:
Please enter the SNMP community string to be used: [Dell_Network_Manager]
Please enter the IP address of the Management System (A.B.C.D) or wildcard (0.0.0.0) to
manage from any Management Station: [0.0.0.0]
```

Enter the following:

* SNMP community string, for example, Dell_Network_Manager.
* IP address of the Management System (A.B.C.D), or wildcard (0.0.0.0) to manage from any Management Station.

Note: IP addresses and masks beginning with zero cannot be used.

Press Enter.

Wizard Step 2

The following is displayed:

```
Now we need to setup your initial privilege (Level 15) user account.
This account is used to login to the CLI and Web interface.
You may setup other accounts and change privilege levels later.
```

```
For more information on setting up user accounts and changing privilege levels, see the
user documentation.

To setup a user account:

Enter the user name<1-20>:[admin]

Please enter the user password:*

Please reenter the user password:*
```

Enter the following:

* User name, for example "admin"
* Password and password confirmation.

Note: If the first and second password entries are not identical, you will be prompted until they are identical.

Press Enter.

Wizard Step 3

The following is displayed:

```
Next, an IP address is setup.

The IP address is defined on the default VLAN (VLAN #1), of which all ports are members.
This is the IP address you use to access the CLI, Web interface, or SNMP interface for
the switch.To setup an IP address:

Please enter the IP address of the device (A.B.C.D):[1.1.1.1]

Please enter the IP subnet mask (A.B.C.D or nn): [255.255.255.0]
```

Enter the IP address and IP subnet mask, for example 1.1.1.1 as the IP address and 255.255.255.0 as the IP subnet mask.

Press Enter

Wizard Step 4

The following is displayed:

```
Finally, setup the default gateway.

Please enter the IP address of the gateway from which this network is reachable (e.g.
192.168.1.1).Default gateway (A.B.C.D):[0.0.0.0]

Enter the default gateway.
```

```
Press Enter. The following is displayed (as per the example parameters described):

This is the configuration information that has been collected:

==============================================================

SNMP Interface = Dell_Network_Manager@0.0.0.0
User Account setup = admin
Password = *
Management IP address = 1.1.1.1 255.255.255.0
Default Gateway = 1.1.1.2

==============================================================
```

Wizard Step 5

The following is displayed:

```
If the information is correct, please select (Y) to save the configuration, and copy to
the start-up configuration file. If the information is incorrect, select (N) to discard
configuration and restart the wizard: (Y/N)[Y]Y
```

Enter [N] to skip to restart the Setup Wizard.

Enter [Y] to complete the Setup Wizard. The following is displayed:

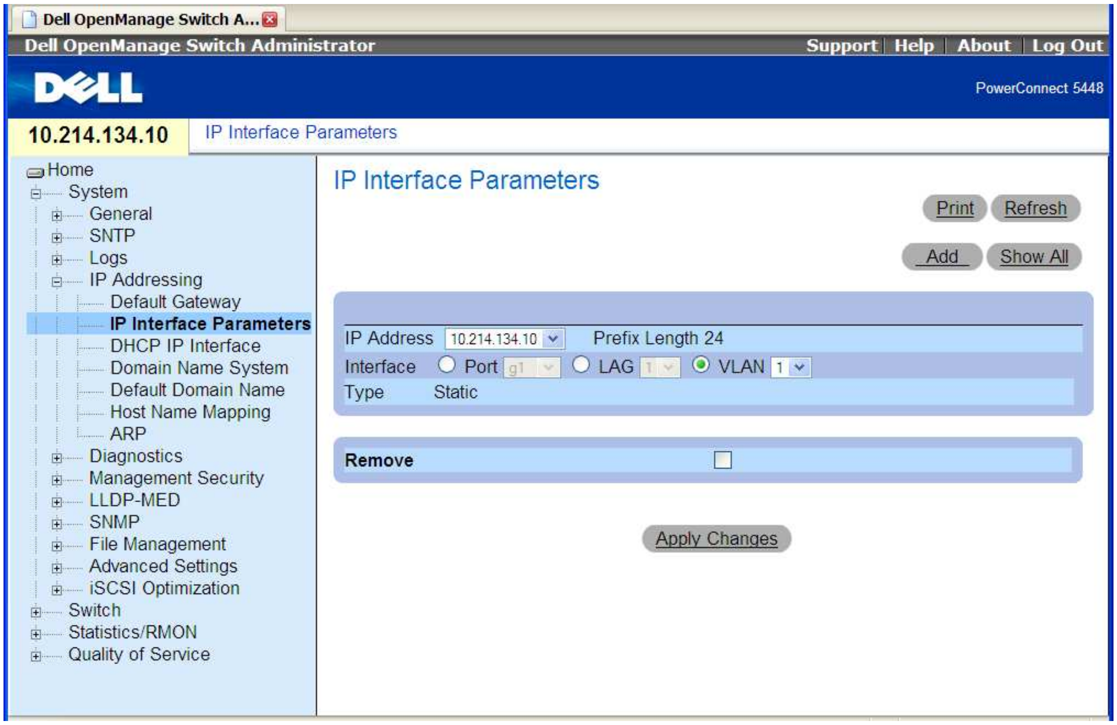
```
Configuring SNMP management interface
Configuring user account......
Configuring IP and subnet......

Thank you for using Dell Easy Setup Wizard. You will now enter CLI mode.
```

Wizard Step 6 -- The CLI prompt is displayed.

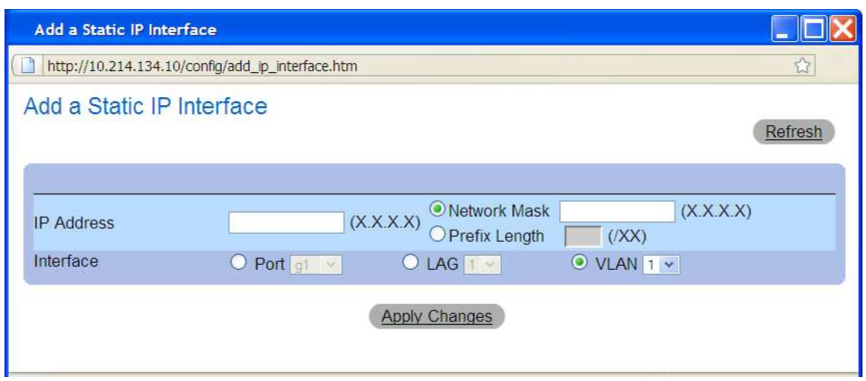
At this point open a web browser and point it to the IP address of the switch. This will present the login for the switch's management GUI. Log in with the 'admin' username and password you supplied during the setup wizard.

To change the IP and subnet mask, select: **System > IP Addressing > IP Interface Parameters**.

Next, you will see the following IP Interface Parameters screen.

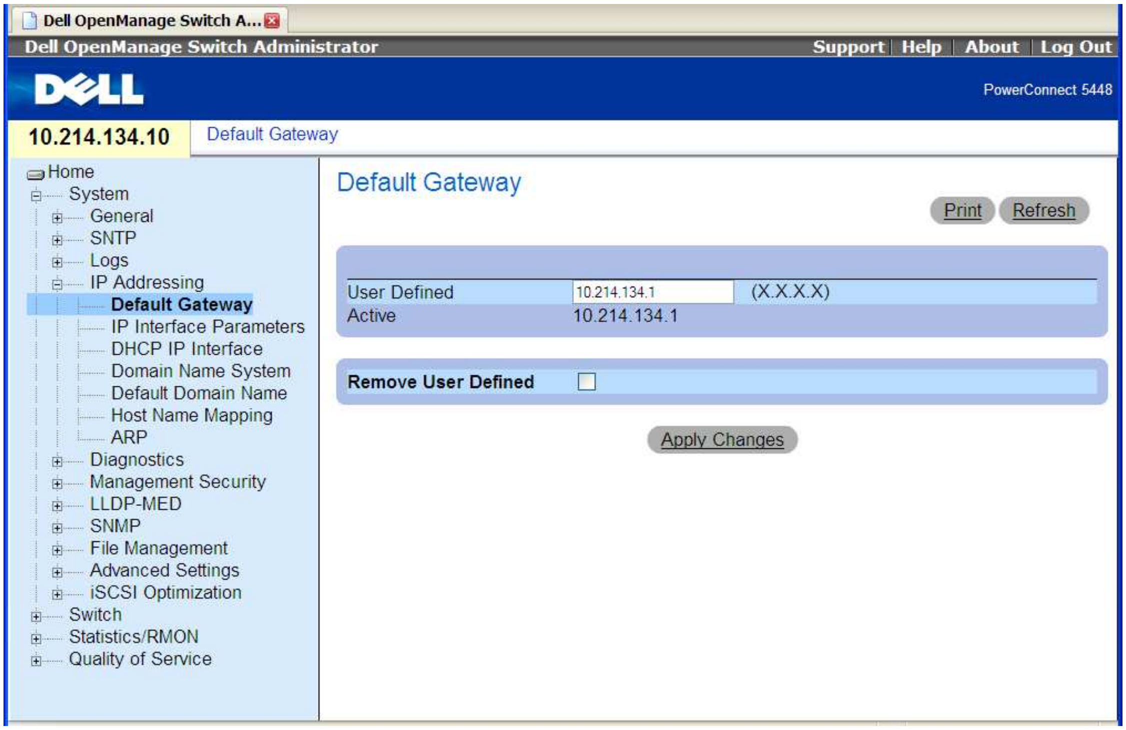


Click **Add** in the upper right corner to add the new customer-provided IP address for the switch and its subnet mask.



Enter the customer-provided IP address for the switch and its subnet mask. Confirm VLAN is selected and 1 is specified for the Interface setting. Click **Apply Changes** to save your changes.

To change the default gateway, click on **System > IP Addressing > Default Gateway**



Enter the customer provided default gateway for their network, then click **Apply Changes** to save.

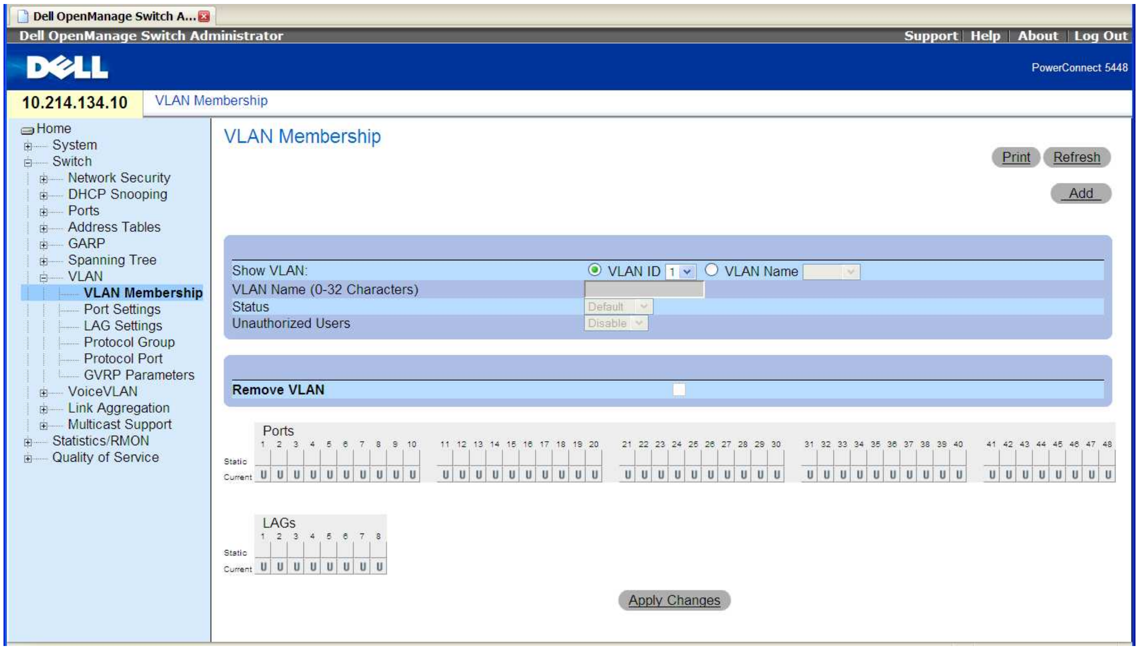
Configuring VLANs for Satori

This procedure will configure the three VLANs for the Satori Server system as follows:

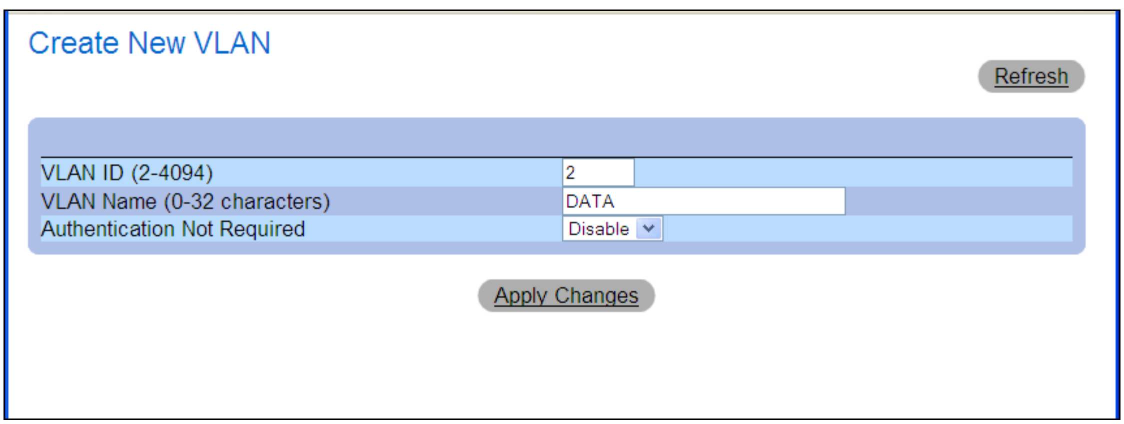
VLAN	Ports
DATA (Satori Backbone)	1 to 35 Odd numbered ports only
HEARTBEAT (Satori control)	2 to 36 Even numbered ports only
CUSTOMER (default vlan)	Remaining ports (37-48)

This VLAN configuration corresponds to the design of the pre-racked Satori solution. Refer to the section Planning for Deployment for information on the requirements for the Satori network.

Next, configure the required VLANs on the switch. Click on **System > IP Addressing > VLAN Membership**. The default configuration is all ports in VLAN 1. This default VLAN 1 will be used as the CUSTOMER VLAN.

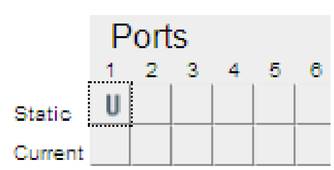


Click the **Add** link in the upper right of this screen. The following popup window will allow you to add the additional required VLANs, DATA and HEARBEAT:

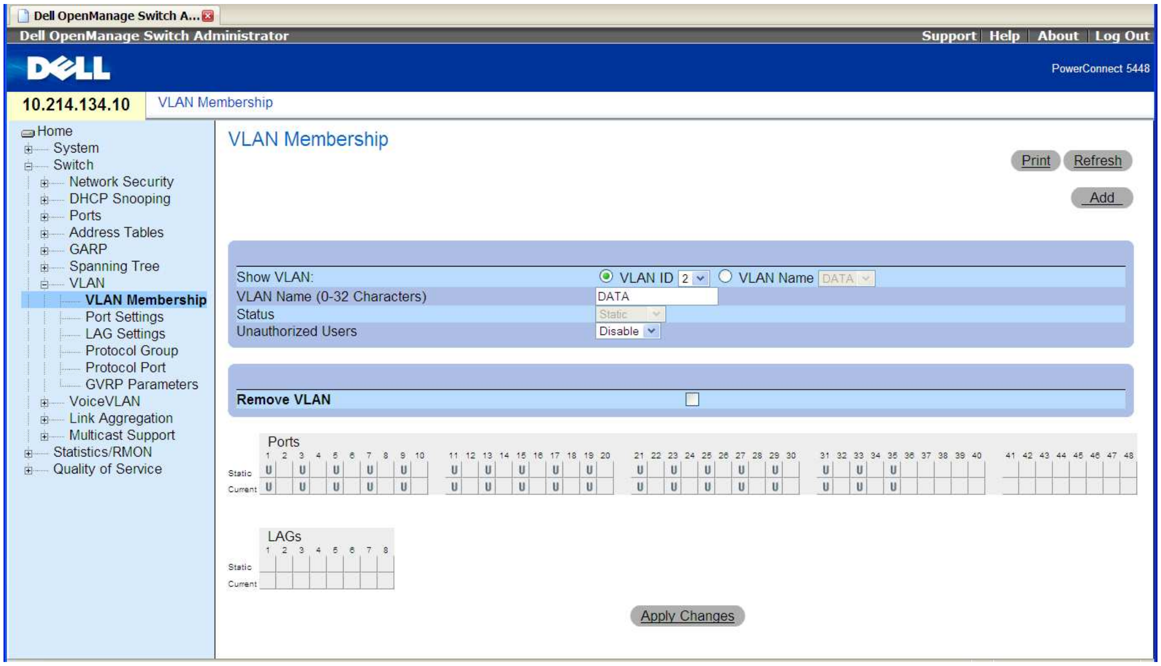


Add the DATA VLAN specifying **2** as the VLAN ID. Click **Apply Changes** to save the VLAN. After the changes have been saved, click the refresh link and add the HEARTBEAT VLAN specifying **3** as the VLAN ID. Again, click **Apply Changes** to save the VLAN. Close the pop-up window after both VLANS have been added.

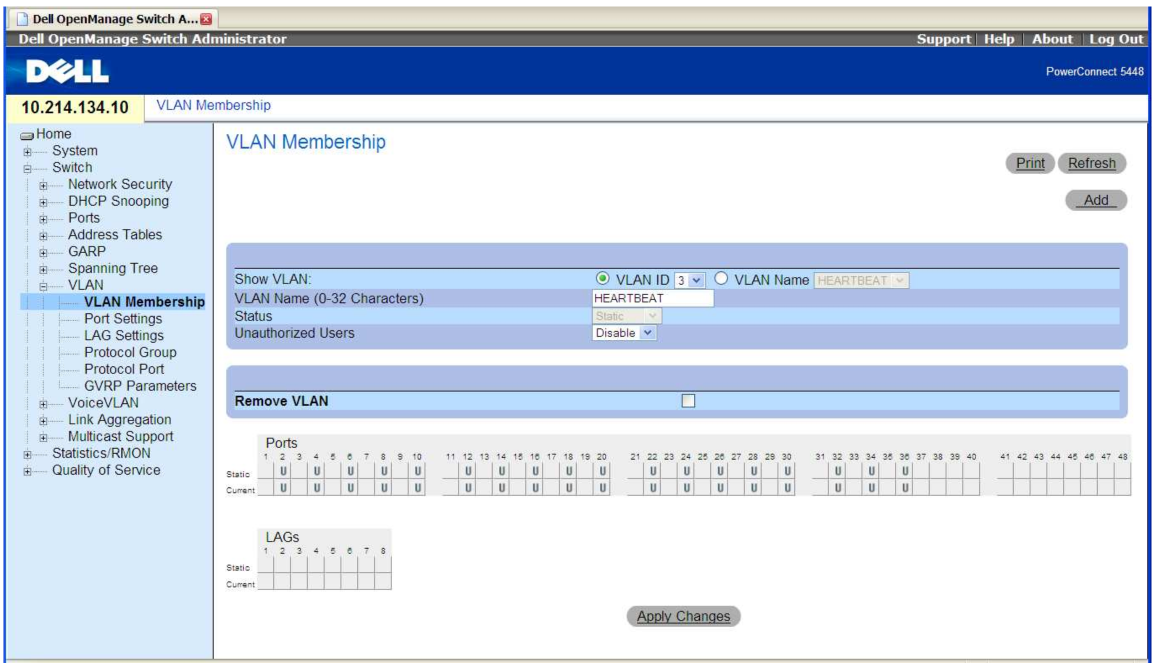
Back on the main VLAN Membership screen, click **Refresh**, highlight VLAN ID, and select **2** from the dropdown list. This will allow you to allocate ports to the DATA VLAN. In the Ports section following the *Static* row, click twice to specify a **U** in the box to add that port to the DATA VLAN. Repeat this for each ODD numbered port from 1 to 35.



Once you have specified a **U** in the Static column for each ODD port from 1 to 35, click **Apply Changes** to save these ports in the DATA VLAN. The resulting DATA VLAN should look like the image below:



Repeat this port allocation process for the HEARBEAT VLAN. Specify EVEN ports 2 to 36 for the HEARBEAT VLAN. The resulting HEARBEAT VLAN should look like this:

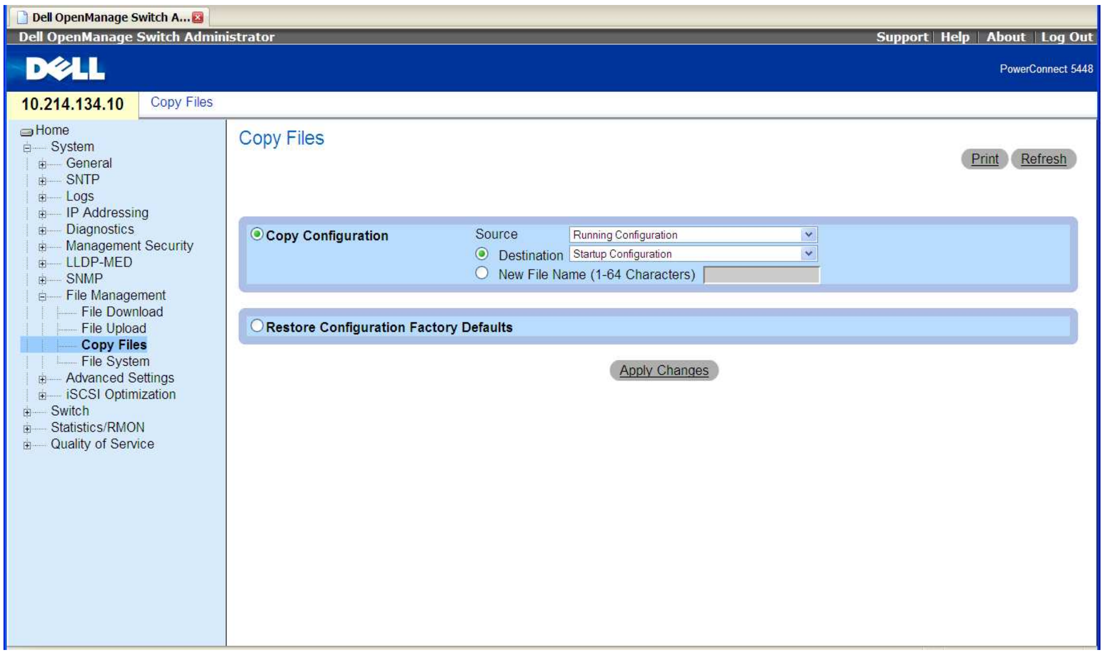


SAVING SWITCH CONFIGURATON

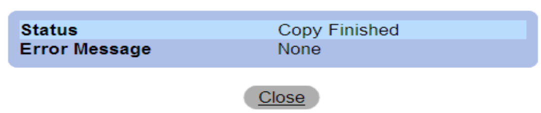
Once the necessary IP addresses and VLANs on the switch have been changed/configured, you need to save the current running configuration to be the startup configuration for the switch. Doing this will ensure the switch will maintain this configuration in the event of a power outage.

To save the running configuration as the startup configuration on the switch, select **System > File Management > Copy files**.

Confirm that the **Copy Configuration** is selected. Select the **Source** as **Running Configuration** and select **Destination** and specify **Startup Configuration**. Click **Apply Changes** to save your changes.



The Copy File operation will provide feedback in the form of a browser pop-up window:



Finally, plug the customer's uplink cable into the switch. You can use any port in the CUSTOMER VLAN, typically port 48. Confirm the changes by sending a ping to the customer-provided default gateway and any other known customer network IP addresses.

Testing the Switch

Verify that the host DB System can see the switch on the network (ping the switch from the host DB).

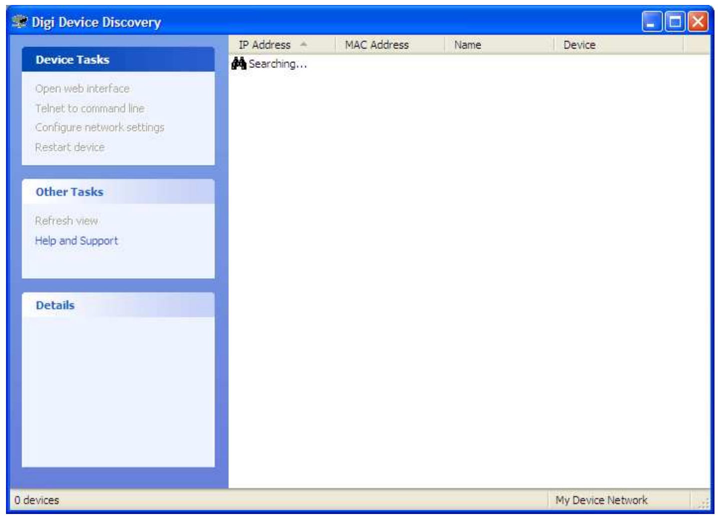
Configure the Terminal Server

Install the DIGI Device Discovery Utility

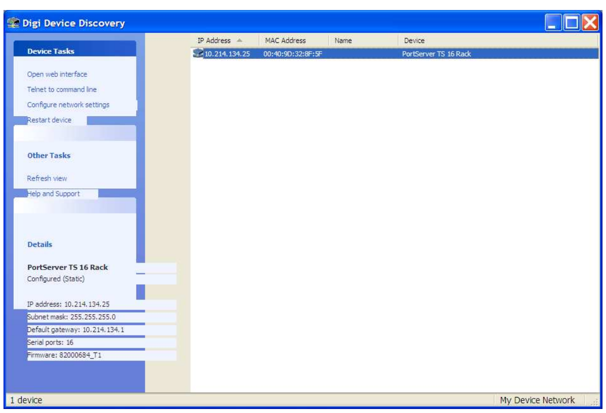
In order to configure the Terminal Server, you must install the DIGI Device Discovery utility. This utility is available on the Dataupia support portal at http://www.dataupia.com. Click **Services > Customer Login**, or click the Customer Login link at the top right of the page.

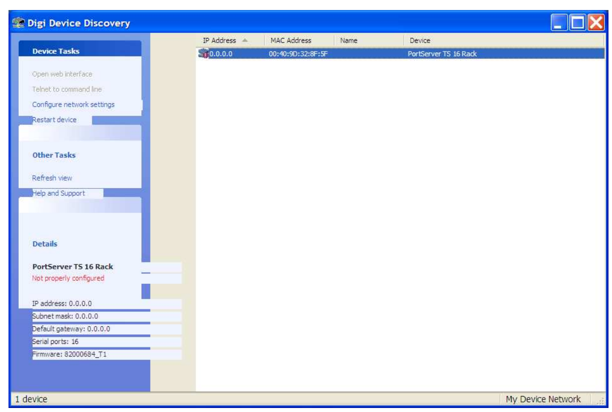
Confirm that the Terminal Server **is** connected to the switch in the CUSTOMER VLAN. Confirm that your laptop **is** connected to the switch and that you have an IP address on the customer's network (it should still be set this way from previous step of configuring the switch). Also, confirm that the uplink to the customer's uplink **IS NOT** connected to the switch. This is to prevent you from "discovering" any devices on the customer's network.

Start the DIGI Device Discovery utility. It will immediately begin searching for devices (the Terminal Server). The screen will look like this (right).

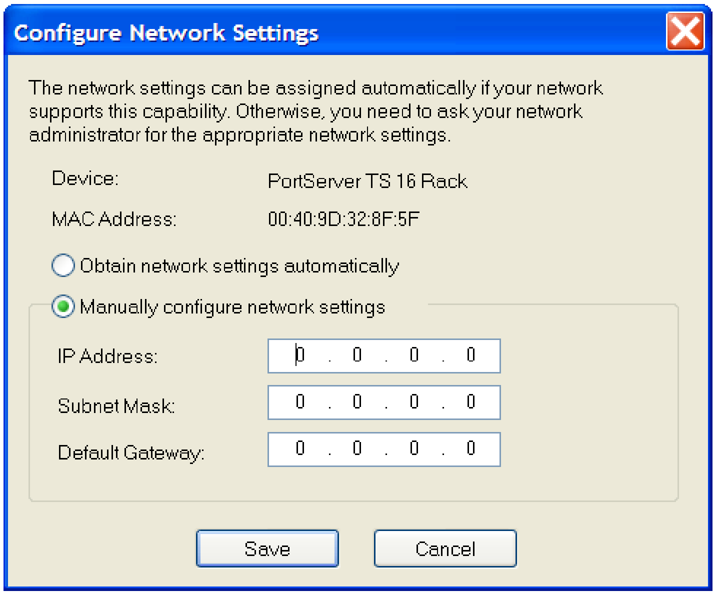


Once the utility locates the Terminal Server, it will appear in the Discovery utility. As shown below, it may already have an IP address configured or it may be set to factory default (0.0.0.0). In either case, configure it with one of the provided IP addresses from the customer's network.



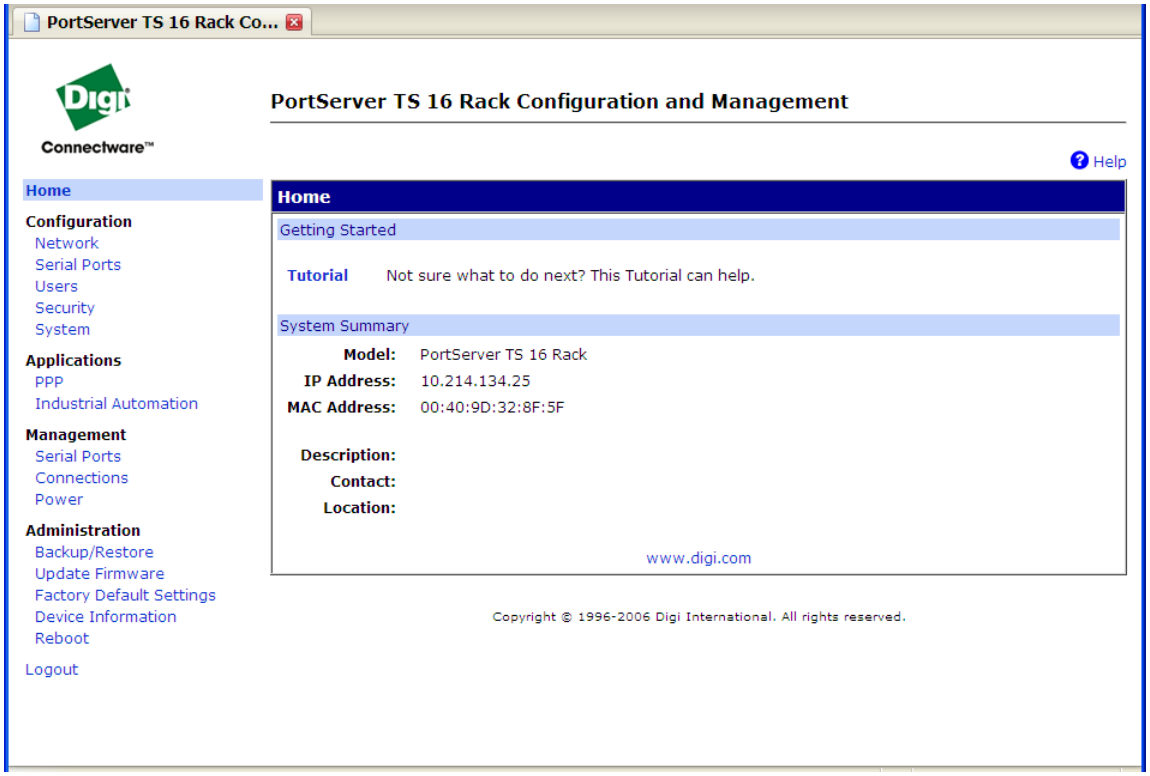


Click **Configure Network Setting** in the left panel (or double-click the device entry) to open the Configure Network Settings window. Select **Manually configure network settings** and type in the IP address, the Subnet Mask, and the Default Gateway provided by the customer. Click **Save** to save the configuration and return to the discovery utility's main window.

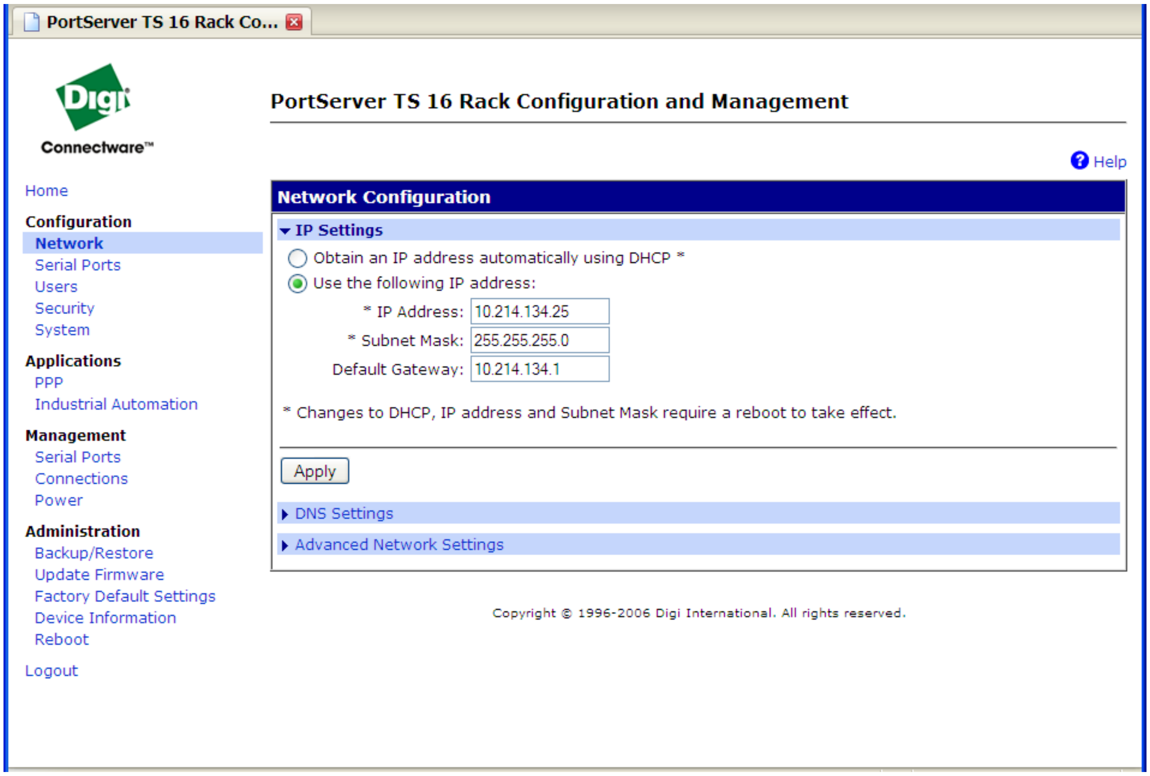


The device's IP address information is now set and you can open the web GUI to complete the configuration of the Terminal Server.

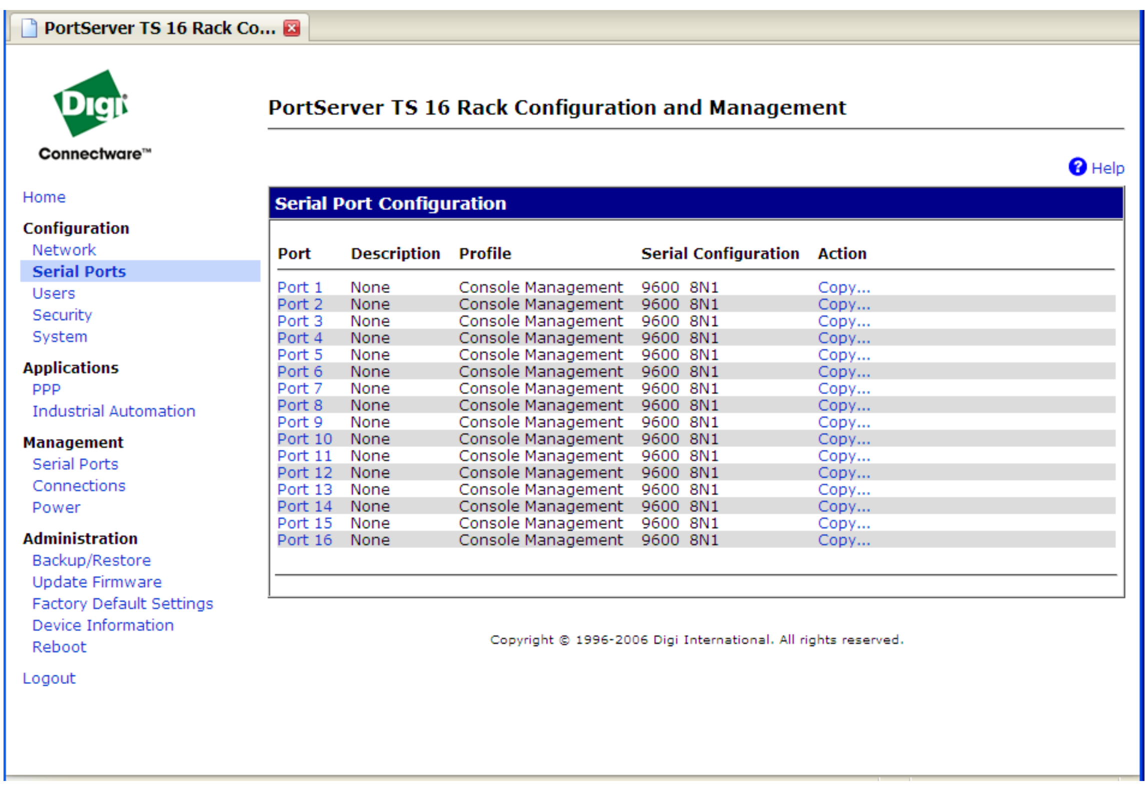
To open the Terminal Server's web GUI, click **Open web interface** from the left panel in the discovery utility's main window. You will be presented with a login screen; use the default credentials of user: root, password: dbps. Once logged in, you will see the following screen.



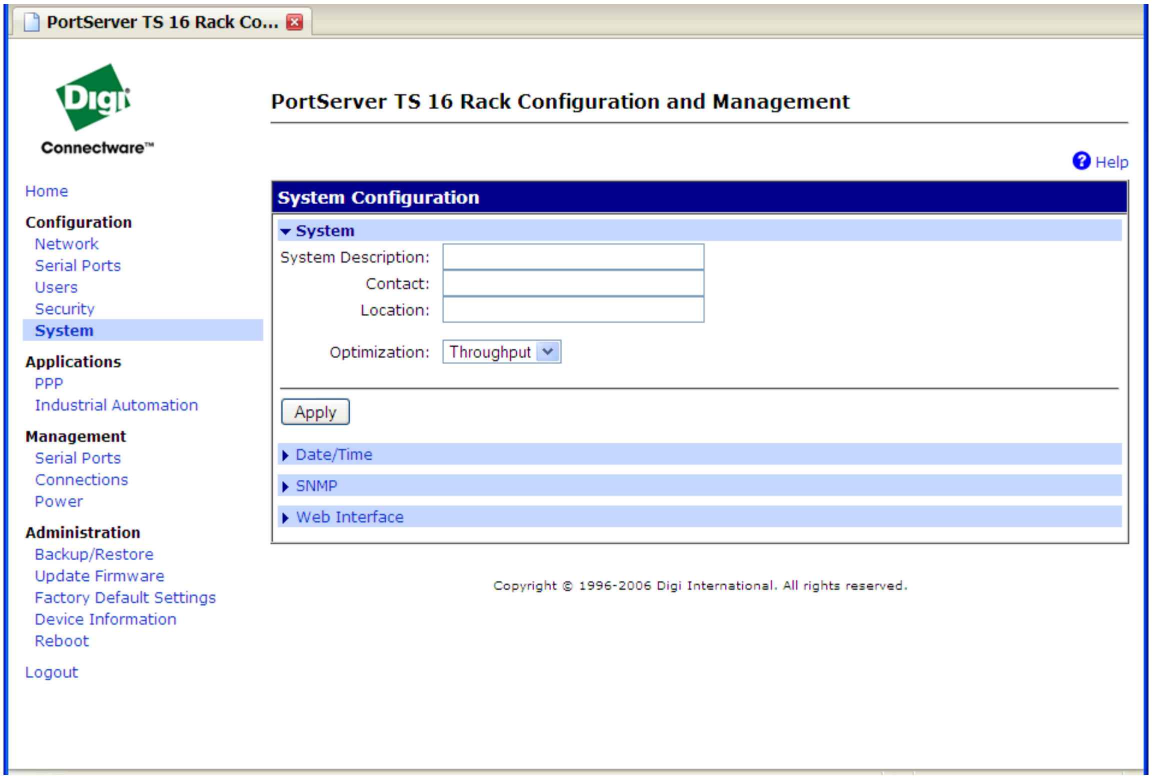
Clicking **Configuration > Network** will allow you to confirm (and if required, change) the IP configuration for the device.



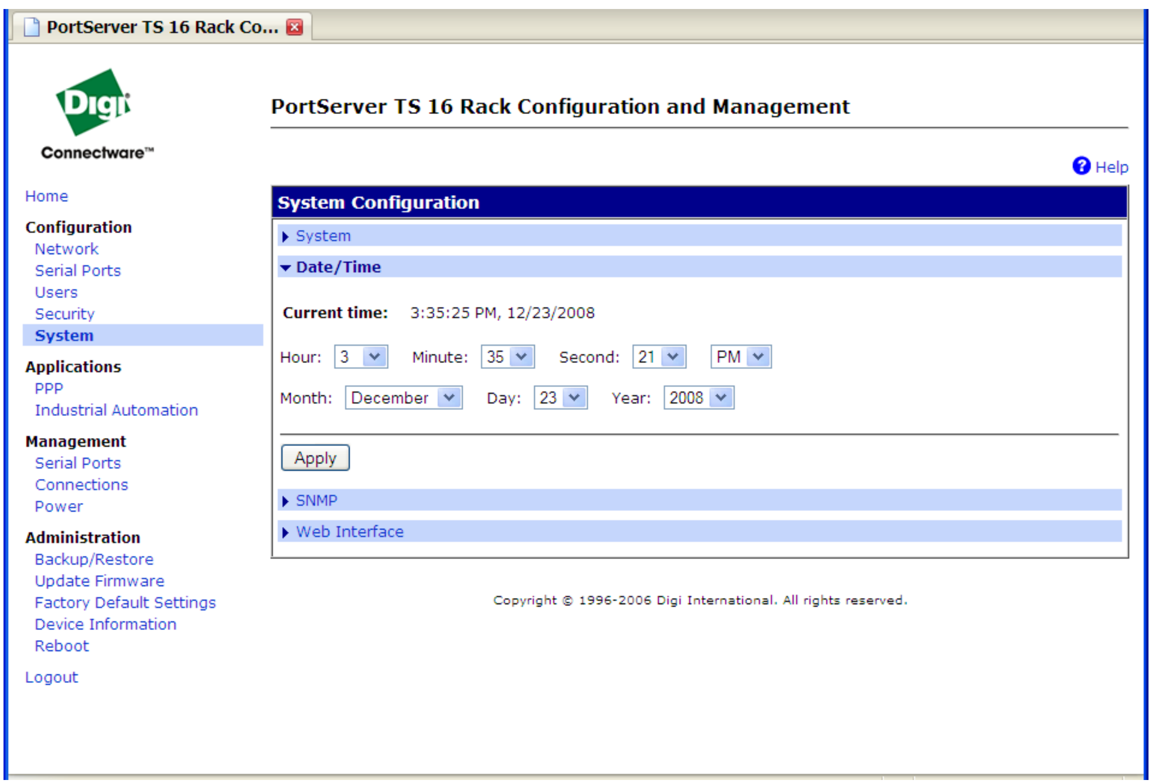
To configure each port in the Terminal Server, click **Configuration > Serial Ports**. You will see the following list of ports. Clicking a link in the **Port** column will allow you to specify a description and change the profile for a port (if necessary).



Additionally, you can provide system-level device information for the Terminal Server by clicking **Configuration > System**. This allows you to provide a description, contact, and location.



On the same System Configuration screen, click **Date/Time** to set the date and time.



Configuring the APC Power Distribution Units (PDUs)

Configure network settings on the APC PDUs:

Configure static IP and other network settings to the Power Distribution Units.

Manufacturer information for the PDUs (by model number) is available on the APC web site:

http://www.apc.com/products/family/index.cfm?id=136

Easiest way to configure the network settings:

You can use a local computer to connect to the PDU to access the control console. You will need the APC serial cable that came with the PDU, which is included the parts kit that comes with a Satori Server pre-racked solution. Normally the cable is plugged into one of the unused terminal server ports using an adapter.

Hint: If you don't have a local computer or laptop with a serial port, you can use a USB serial converter (or a port on the terminal server) and telnet to it from one of the Satori Servers using the KVM or a local keyboard and terminal.

1. Select a serial port at the local computer, and disable any service which uses that port.
2. Use the serial cable (APC PN 940-0144) to connect the selected port to the serial port on the front panel of the PDU. The serial cable ships with the PDU, and is located in the parts kit.
3. Run a terminal program (such as HyperTerminal®) and configure the selected port for 9600 bps, 8 data bits, no parity, 1 stop bit, no flow control, and save the changes.
4. Press ENTER to display the User Name prompt.
5. Use apc for the user name and password.
6. Choose Network from the Control Console menu.
7. Choose TCP/IP from the Network menu.
8. Select the Boot Mode menu. Select Manual boot mode, and then press ESC to return to the TCP/IP menu. Set the System IP, Subnet Mask, and Default Gateway address values. (Changes will take effect when you log off.)
9. Press CTRL+C to exit to the Control Console menu.
10. Log off (option 4 in the Control Console menu).

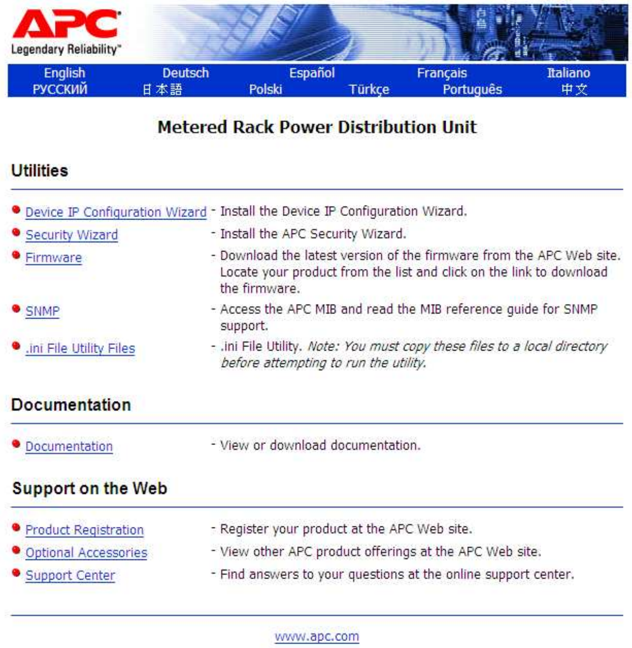
If you don't have a serial cable, use this alternate method to connect to the PDU to configure it.

First, install the APC Device IP Configuration Wizard. This is available on the Dataupia Support Portal at http://www.dataupia.com. Click **Services > Customer Login**, or click the Customer Login link at the top right of the page.

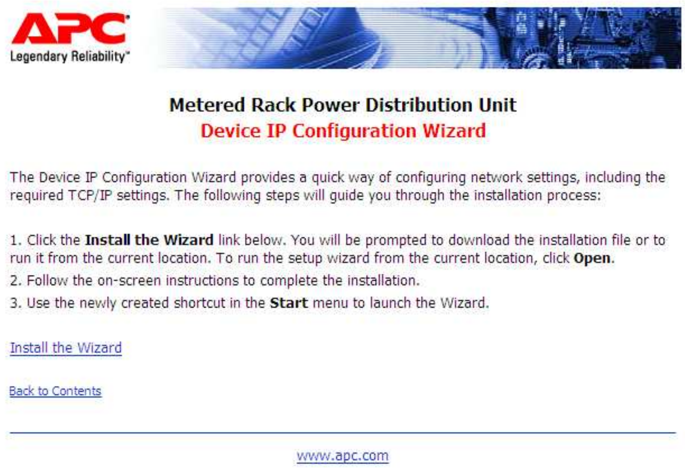
Confirm that the PDUs are connected to the switch in the CUSTOMER VLAN. Confirm that your laptop is connected to the switch and you have an IP address on the customer's network (it should still be set this way from previous steps). Also, confirm that the uplink to the customer's uplink IS NOT connected to the switch. (This is to prevent you from "discovering" any devices on the customer's network.)

Locate and insert the APC Switched Rack Power Distribution Unit *Utility CD* into your laptop. Once inserted, the CD should autoplay and open in a browser; if not, explore the CD and open the file **contents.htm** in a browser. You will only need to perform the steps on this page once. The same Wizard will be used to configure an IP address on each of the new PDUs.

You will see the page to the right (or one very similar to it) in your browser. Click **Device IP Configuration Wizard** under **Utilities**.



On the next page, click **Install the Wizard** to copy the Wizard software to your laptop.



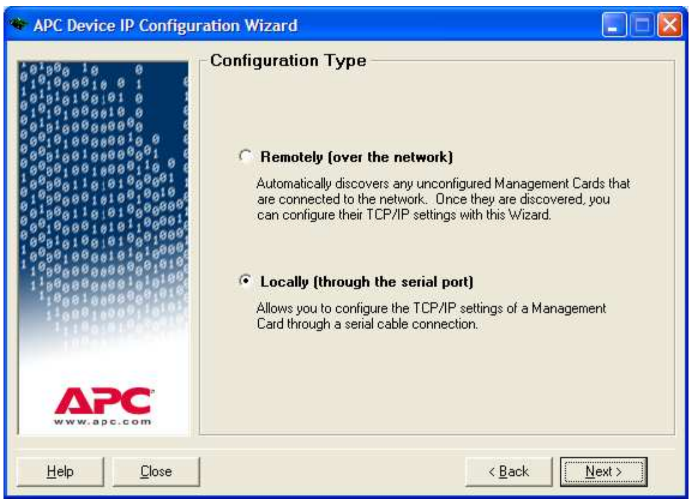
Click **Save File** to save the Install Wizard to your laptop. It may ask you a location, anywhere you like including your Desktop is fine.

The file **APC Device IP Configuration Wizard.exe** (or something very similar to it if there's a newer version; watch the window for the actual name) will be copied to your laptop. Locate this file and run it to install the Device IP configuration Wizard.

At this point, you should be plugged into the first PDU you want to configure. Use the cable mentioned previously; the one to connect your laptop's serial connection to the RJ11 port on the PDU. It should be the port on the LEFT looking at the unit. The port on the right should be the RJ45 port.

When the install completes, it should launch the Wizard and you will see this screen (right).

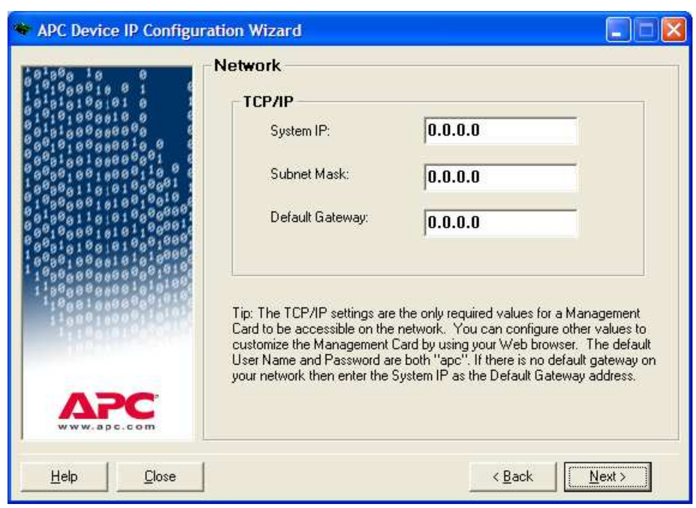
The Wizard may automatically discover the PDU you are connected to and jump to the third screen (at the bottom on this page) but if not, click **Next**.

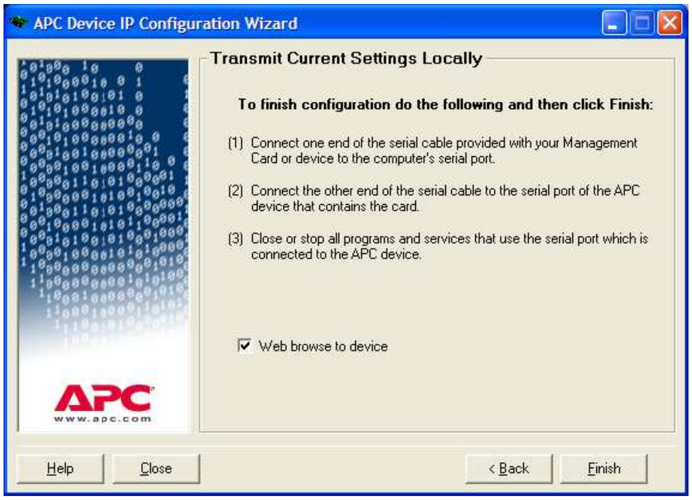


On the Configuration Type screen, select **Locally (through the serial port)**, then click **Next**.

On the Network screen, provide the System IP (listed previously in “Configure PDUs with IP address”) for the IP address of the PDU you are connected to; the Subnet Mask is 255.255.255.0, and the Default Gateway is 172.21.100.254

Click **Next**.



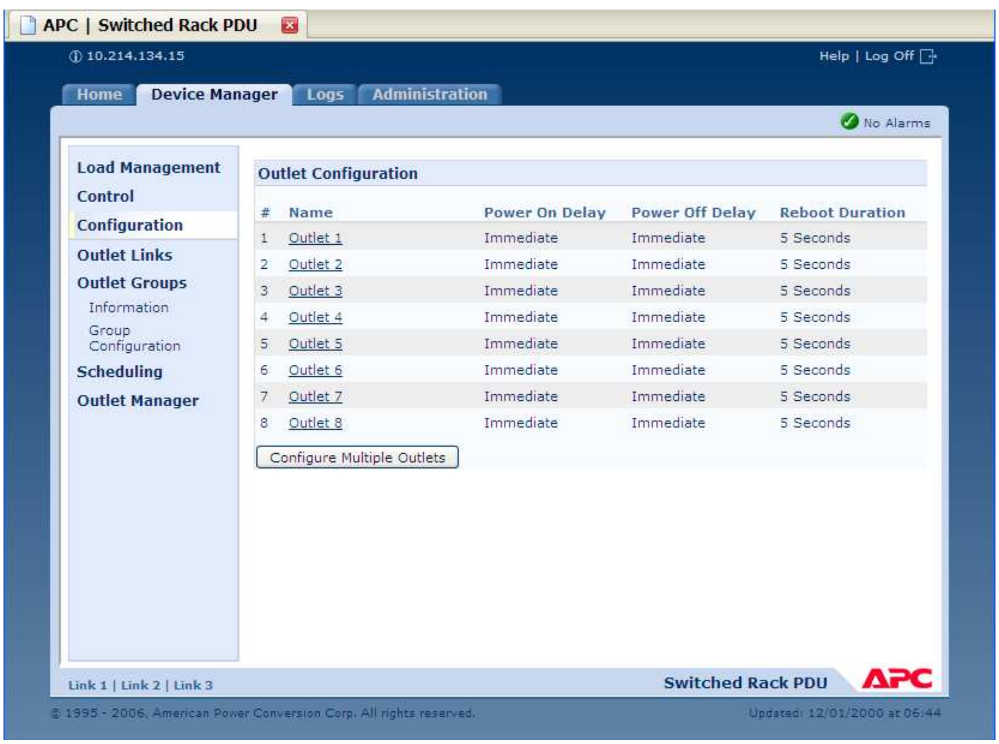


Confirm that you have entered the IP information for the PDU you are actually connected to and click **Finish**.

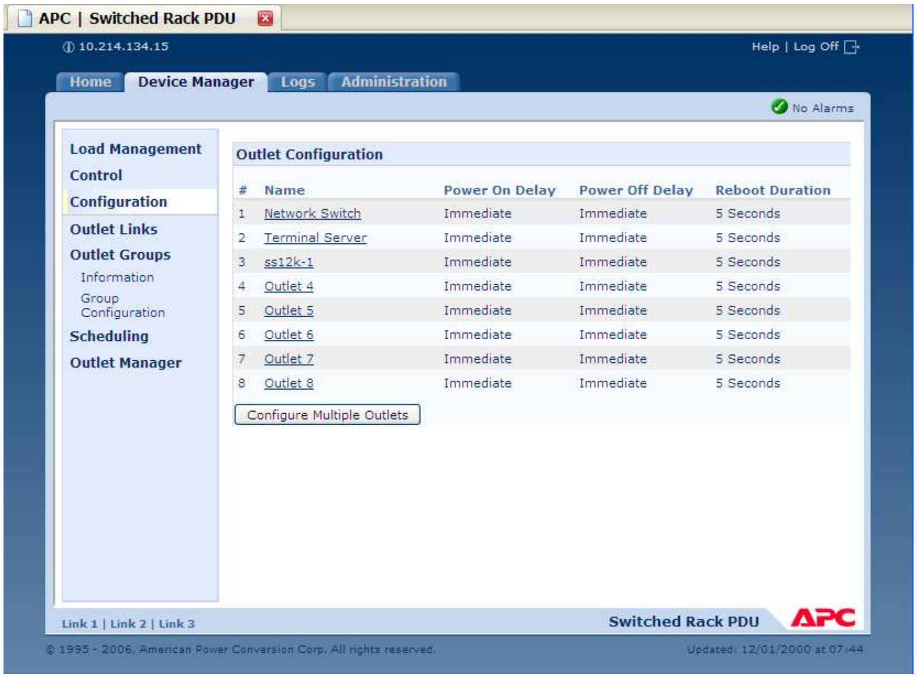
Your browser should open and (if your laptop is on the same IP range as the device) point to the configuration page of the PDU.

Note: These steps may vary slightly with newer versions of the Wizard.

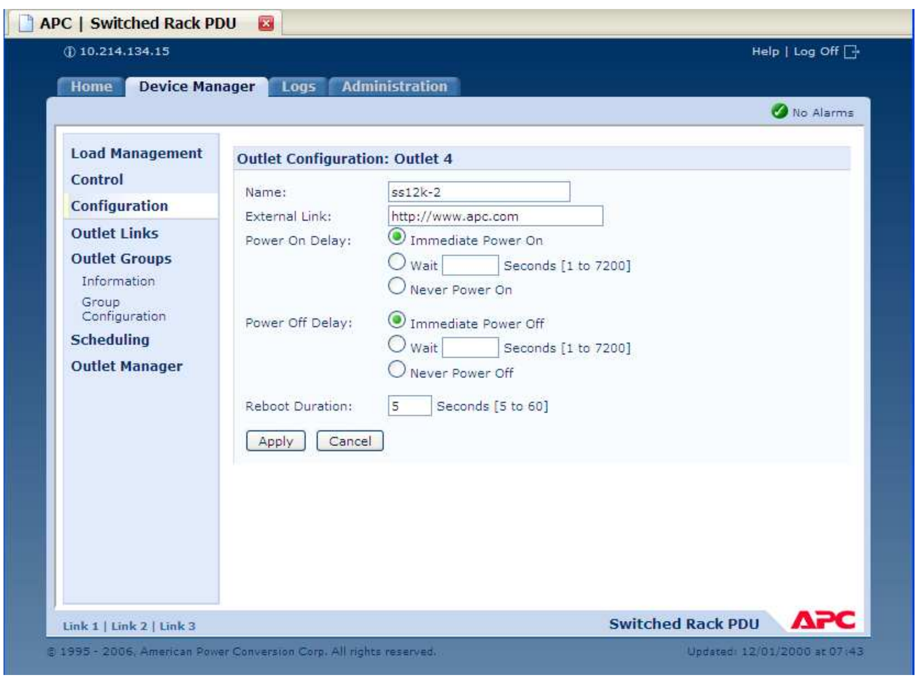
Click the **Device Manager >Configuration** tab to display the default outlet configuration.



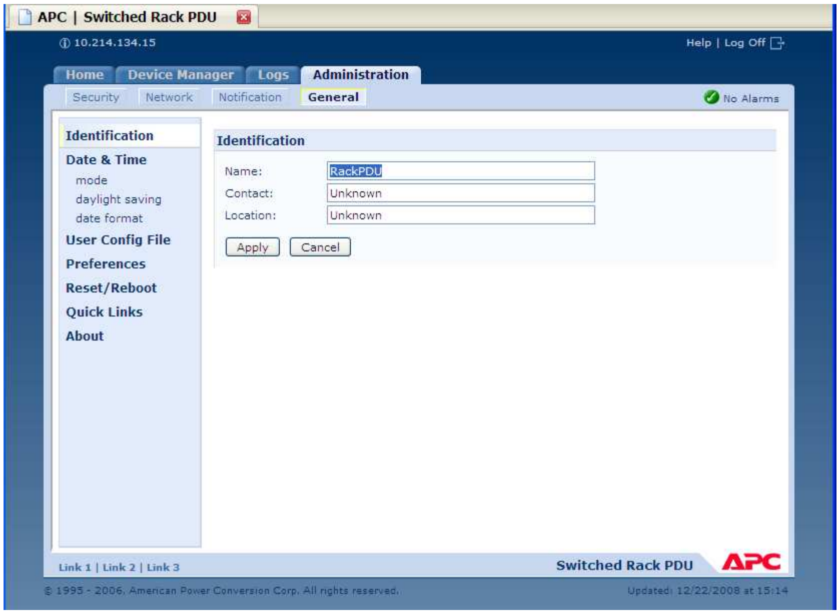
Clicking on the name/link for an outlet will allow you to provide information specific to that outlet, as well as to configure behavior.



In the **Name** field, type in the name of the device connected to that outlet.



You can use the **Administration** tab to configure general device settings.

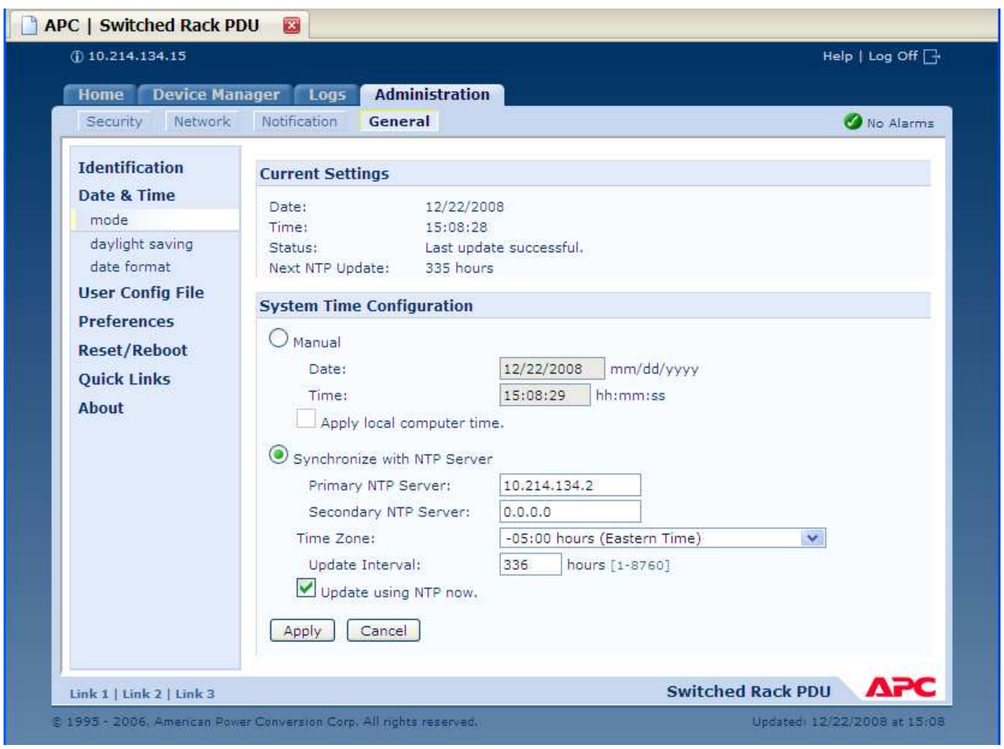


Select the **General** subtab to configure device information.

In the left panel, select **Identification** to specify device name, contact and location information.

Click **Apply** to save.

To configure date and time information, select **mode** under **Date & Time** in the left panel. Here you can configure the PDU to point to a NTP server for system time synchronization.



Select **Synchronize with NTP Server** and provide the NTP Server's IP address. Select the appropriate Time Zone and select **Update using NTP now**. Click **Apply** to save the changes.

At this point, you should reconnect the customer's uplink cable to the switch. This will connect the devices you have just configured, the switch itself, the Terminal Server, and the PDUs to the CUSTOMER VLAN, and thus to the customer's network. Now, the customer should be able to ping the switch you set up -- as well as the Terminal Server and the PDUs -- from their network. You should retain the IP addresses you assigned to these devices and provide them to Customer Support.

Configure Peripherals

Configure the Loader Blade or DA Blade as an NTP Server

To configure time synchronization on a Satori solution, select a Loader or a DA (Dynamic Aggregation Server) blade to be the single NTP server for the solution and have all blades (and other Loaders and DAs) reference it. Then "point" the Loader (or DA) to a customer provided NTP server as an authoritative NTP source. The steps to configure this follow.

Steps for configuring a LOADER (or DA) blade as an NTP Server

Acquire, from the customer, the server name or IP address of the NTP server they want you to "point" the Satori Solution to for time synchronization.

Confirm that the IP address provides the NTP service and is reachable from the Loader (or DA):

```
ntpdate <server_name/IP>
```

Configure the ntpd with the provided NTP server information from the customer. Edit /etc/ntp.conf and make the following changes: use a # symbol to comment out references to other servers, and add the server entry for the provided server.

--LIKELY ORIGINAL--

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool
(http://www.pool.ntp.org/join.html)
server 0.fedora.pool.ntp.org dynamic
server 1.fedora.pool.ntp.org dynamic
server 2.fedora.pool.ntp.org dynamic
```

--CHANGE TO-- (changes in **red**)

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool
(http://www.pool.ntp.org/join.html)
#server 0.fedora.pool.ntp.org dynamic
#server 1.fedora.pool.ntp.org dynamic
#server 2.fedora.pool.ntp.org dynamic
server <server_name/IP> dynamic
```

--EXAMPLES--

```
server sundial.columbia.edu dynamic
```

```
server 128.59.59.177 dynamic
```

Synchronize the system time to the hardware clock on the Loader/DA so the time remains true:

```
hwclock --systohc
```

Make the system time and hardware clock synchronization persistent by configuring the change in /etc/sysconfig/ntpd.

--LIKELY ORIGINAL--

```
# Set to 'yes' to sync hw clock after successful ntpdate
SYNC_HWCLOCK=no
```

--CHANGE TO-- (change in **red**)

```
# Set to 'yes' to sync hw clock after successful ntpdate
SYNC_HWCLOCK=yes
```

Configure the NTP daemon to start on boot for the correct run levels.

Check what run levels currently start ntpd `chkconfig --list ntpd`

```
[root@loader ~]# chkconfig --list ntpd
ntpd            0:off   1:off   2:off   3:off   4:off   5:off   6:off
```

Configure NTP to start on reboot `chkconfig --level 35 ntpd on`

(This will start NTP in run levels 3 & 5)

```
[root@loader ~]# chkconfig --level 35 ntpd on
[root@loader ~]# chkconfig --list ntpd
ntpd            0:off   1:off   2:off   3:on    4:off   5:on    6:off
```

Start the NTP daemon on the Loader/DA `/etc/init.d/ntpd start`

```
[root@loader etc]# /etc/init.d/ntpd start
Starting ntpd: [  OK  ]
```

Steps for configuring a Satori Blade to point to an NTP Server

To configure each Satori Server to get its time from a Loader or DA blade, follow these steps:

(You must be in Configure mode on the blade to change NTP settings.)

Check the current NTP configuration on the blade:

```
ss12k-blade-1 (config) # show ntp
NTP enabled: no
No NTP peers configured.
No NTP servers configured.
```

Enable NTP on the blade (is needed):

```
ss12k-blade-1 (config) # ntp enable
```

Add the Loader (or DA) blade as the NTP server; specify the IP address of the server:

```
ss12k-blade-1 (config) # ntp server 10.0.2.20 .....{the Loader or DA's IP}
```

Confirm the changes you made:

```
ss12k-blade-1 (config) # show ntp
NTP enabled: yes
No NTP peers configured.
NTP server 10.214.128.77
  Enabled: yes
  NTP version: 4
```

Persist the changes on the blade:

```
ss12k-blade-1 (config) # write memory
```

To perform a one-time NTP sync with a known NTP server, use the following command:

```
ss12k-blade-1 (config) # ntpdate 10.0.2.20.....{the Loader or DA's IP}
```

Note: NTP must be disabled on the server in order to run this command. If NTP is enabled, an error message will appear (“the NTP socket is in use, exiting”).

Communication Tests (between host DB & Peripherals)

Use the ping command to confirm network connectivity between the following components:

- Host database to peripherals
- Host database to peripherals via SSH/Telnet
- Each peripheral to all other peripherals
- Each peripheral to all other peripherals via SSH/Telnet

Configure the Satori Server Array Network Settings

The only changes on the blade side are to eth0 on the head blade. Configure eth0 on the head blade to a customer assigned IP in the CUSTOMER VLAN. Also, confirm eth0 IS connected to the CUSTOMER VLAN, (green section, ports 36 - 48). You will also need to specify a default gateway on this blade to allow http traffic back to the user's browser on the customer's network.

logon to the array's head blade

`> en`

`# configure terminal`

IP settings for eth0

`(config) # show interfaces eth0` (to see current setting of eth0)

`(config) # interface eth0 ip address {new_IP_address} {new_subnet}`

`(config) # write memory` (to save changes)

`(config) # show interfaces eth0` (to confirm new setting of eth0)

Setting Default Gateway for head blade

`(config) # ip default-gateway {default_gateway}`

`(config) # write memory` (to save changes)

`(config) # show ip default-gateway`

Communication Tests (between host DB & Peripherals)

Use the ping command to confirm network connectivity between the following components:

- Each blade to all other blades
- Each blade to all other blades via SSH/Telnet

Configure the Blades into a Dataupia Array

This procedure is included in the applicable *Dataupia Satori Server Installation and Configuration Guide* (Oracle or MS SQL Server).

Install and Configure the Dataupia Client Package

The Dataupia Client Package includes:

- The Dataupia Connector — Enables the host database server(s) to communicate with one or more Dataupia arrays.
- The Dataupia Data Loader — Loads data from files into tables on a Dataupia array.

To install and configure the Dataupia Client Package on an Oracle database, follow the procedures in the *Dataupia Satori Server Installation and Configuration Guide – Oracle.*

To install and configure the Dataupia Client Package on a MS SQL Server database, follow the procedures in the *Dataupia Satori Server Installation and Configuration Guide – MS SQL Server.*

Appendix A: APC Rack Assembly

If your Satori Server system was not preassembled at Dataupia, and does not have the mounting hardware installed, you can use the following procedures to assemble an APC rack.

11. **Important:** you should remove the rack from the pallet before installing the blades. The rack is much heavier with the blades installed, so it is much more difficult to safely remove the rack from the pallet with the blades installed. Damage to the system or personal injury could result.

Unpacking a 24U APC Rack

Figure 9: A 24U APC Rack

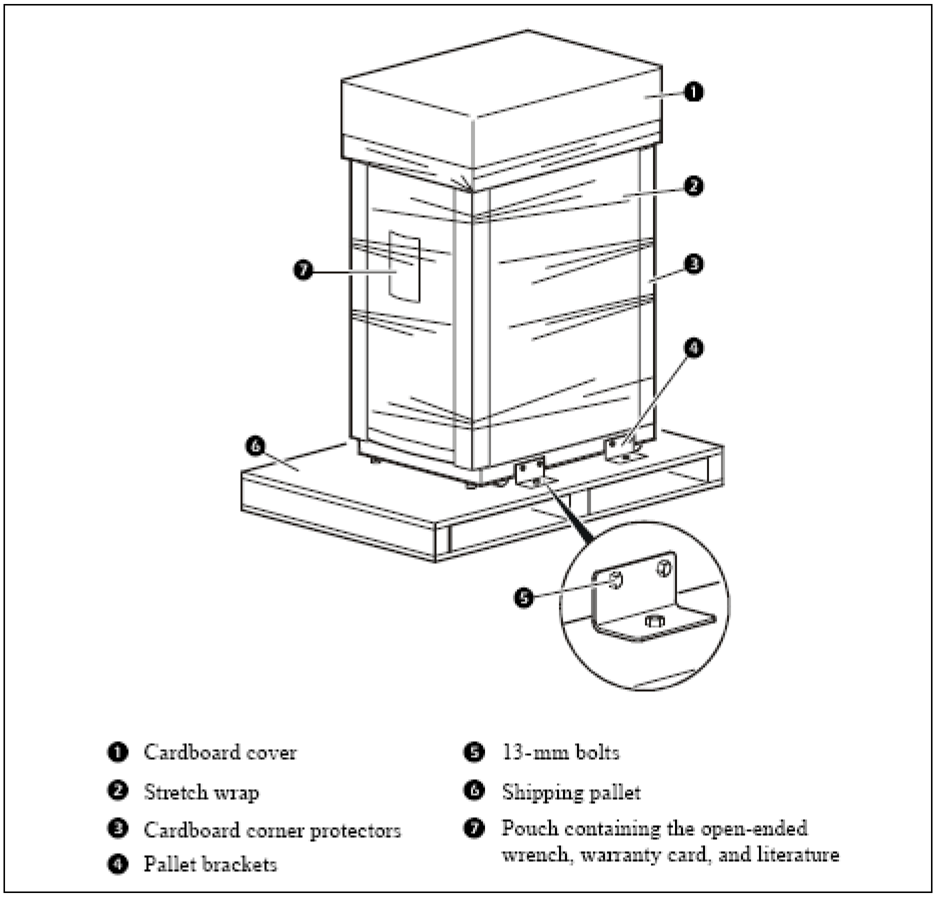


Figure 10: 24U APC Rack Packing Components

How to unpack an APC 24U enclosure

- Move the shipping pallet to a firm, level surface in an open area.
- Remove the open-end wrench from the pouch on the front of the packaging.
- Refer to the label on the packaging to determine where to cut the wrapping. Using sturdy shears or a utility knife, carefully remove the plastic stretch wrap.
- Remove the cardboard cover and the four cardboard corner protectors.

How to remove an empty APC 24U enclosure from its shipping pallet

12. **Important:** you should remove the rack from the pallet before installing the blades. The rack is much heavier with the blades installed, so it would be much more difficult to safely remove the rack from the pallet with the blades installed. Damage to the system or personal injury could result.

    It is highly recommended that you **remove the rack from the pallet before installing the blades**.

1. Remove the four brackets that anchor the enclosure to the shipping pallet. Use the 13-millimeter end of the open-end wrench to remove the bolts that secure the brackets to the pallet.
2. With one person on each side of the enclosure, carefully roll it toward the rear of the pallet until the rear casters clear the back edge of the pallet. Continue to slide the enclosure rearward until the rear casters touch the floor.

3. While one person carefully tips the enclosure slightly away from the pallet, have the other person pull the pallet away from the enclosure.

How to remove the side panels from the rack

To remove the side panels from the rack, press down the two recessed release buttons near the top of each side panel. Pull the top of the panel out and lift the panel off of the rack base.

How to level the rack

Leveling feet are attached under the enclosure at the corners. The leveling feet help provide a stable base if the floor is uneven, but they cannot compensate for an extremely sloped surface.

1. Move the enclosure to a level location.
2. For each leveling foot, fit the 14-millimeter end of the open-end wrench (provided) to the hex head just above the round pad on the bottom of the foot. Turn the wrench clockwise to extend the leveling foot until it makes firm contact with the floor.
3. Use a level to determine which feet need further adjustment.

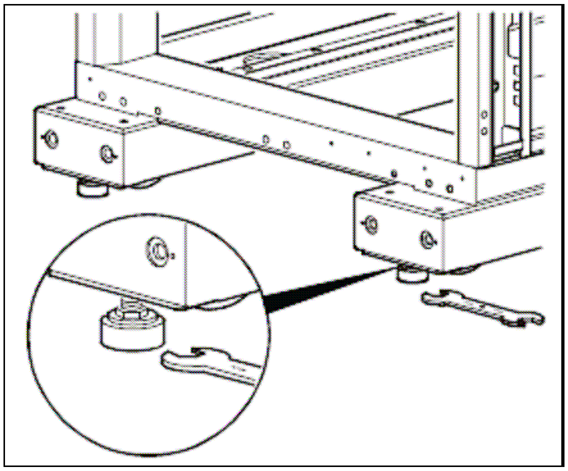


Figure 11: Leveling an APC Rack

For more information about other rack features, please refer to the APC product manual: http://www.apcmedia.com/salestools/ASTE-6Z6K54_R0_EN.pdf.

Unpacking a 42U APC Rack

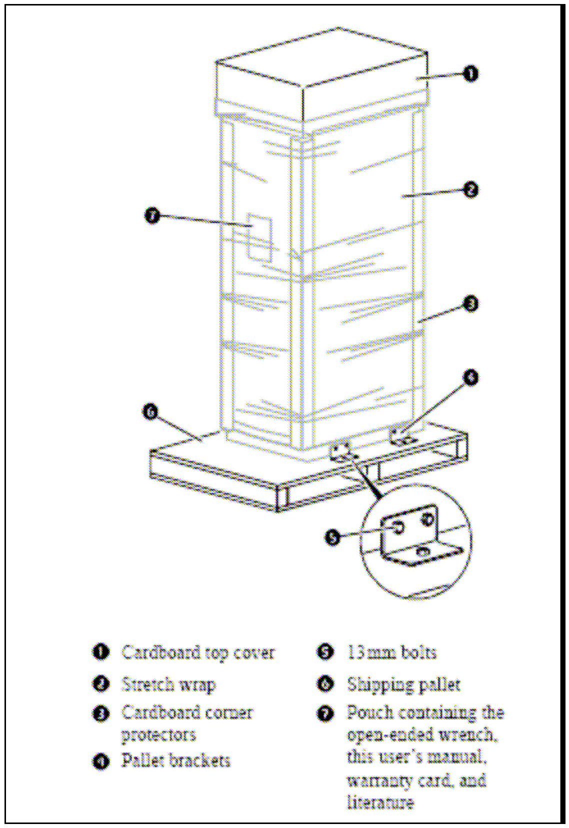


Figure 12: 42U APC Rack Packing Components

How to unpack an APC 42U enclosure

1. Move the shipping pallet to a firm, level surface in an open area.
2. Remove the open-end wrench from the pouch on the front of the packaging.
3. Refer to the label on the packaging to determine where to cut the wrapping. Using sturdy shears or a utility knife, carefully remove the plastic stretch wrap.
4. Remove the cardboard cover and the four cardboard corner protectors.

How to remove an empty APC 42U enclosure from its shipping pallet

13. **Important:** you should remove the rack from the pallet <u>before</u> installing the blades. The rack is much heavier with the blades installed, so it would be much more difficult to safely remove the rack from the pallet with the blades installed. Damage to the system or personal injury could result.

    It is highly recommended that you **remove the rack from the pallet before installing the blades**.

1. Remove the four brackets that anchor the enclosure to the shipping pallet. Use the 13-millimeter end of the open-end wrench to remove the bolts that secure the brackets to the pallet.

2. With one person on each side of the enclosure, carefully roll it toward the rear of the pallet until the rear casters clear the back edge of the pallet. Continue to slide the enclosure rearward until the rear casters touch the floor.
3. While one person carefully tips the enclosure slightly away from the pallet, have the other person pull the pallet away from the enclosure.

How to remove the side panels from the rack

There are two side panel sections on each side of the rack; one on the top, and one on the bottom.

To remove the side panels from the rack, press down the recessed release button near the top of each side panel. Pull the top of the panel out and lift the panel off of the rack.

How to level the rack

Leveling feet are attached under the enclosure at the corners. The leveling feet help provide a stable base if the floor is uneven, but they cannot compensate for an extremely sloped surface.

4. Move the enclosure to a level location.
5. For each leveling foot, fit the 14-millimeter end of the open-end wrench (provided) to the hex head just above the round pad on the bottom of the foot. Turn the wrench clockwise to extend the leveling foot until it makes firm contact with the floor.
6. Use a level to determine which feet need further adjustment.

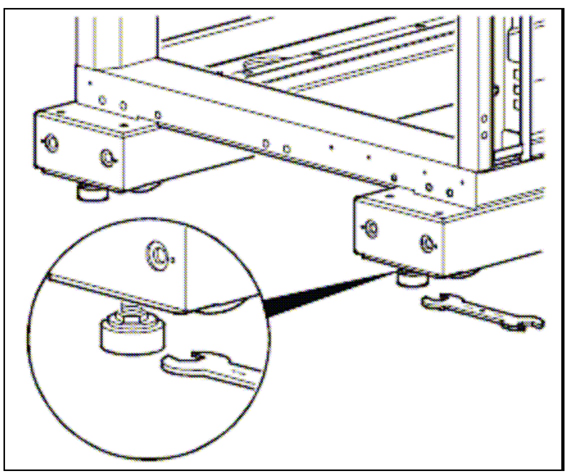


Figure 13: Leveling an APC Rack

For more information about other rack features, please refer to the APC product manual: http://www.apcmedia.com/salestools/ASTE-6Z6K8X_R0_EN.pdf.

Installing PDU Mounting Hardware

Two different types of Power Distribution Units (PDUs) are installed in Dataupia systems. In the 24U Rack, rack-mounted PDUs are usually installed at the bottom of the rack. In the 42U Rack, vertical PDUs are installed at the back of the rack.

Installing PDU Mounting Hardware in a 24U APC rack

If you have not already done so, remove the side panels from the rack.

When installing equipment, locate the top and bottom of a U-space on the mounting rails. Each U-space is indicated by white horizontal marking lines, and contains three square mounting holes. The number of each U-space appears next to the middle mounting hole.

The PDUs in the 24U APC rack are secured with M6 caged nuts inserted into the top and bottom mounting holes in U-spaces 1 and 2 (the first two U-spaces at the bottom of the rack) at each of the four vertical mounting rails:

Figure 14: PDU Mounting Hardware in a 24U APC Rack

How to install and remove caged nuts

Note: install the caged nut horizontally with the sides of the caged nut engaging the left and right sides of the square hole.

1. Insert the caged nut into the square hole by hooking one side of the caged nut assembly through the far side of the hole.
2. Place the caged nut tool (provided) on the other side of the caged nut and pull to snap it into position.

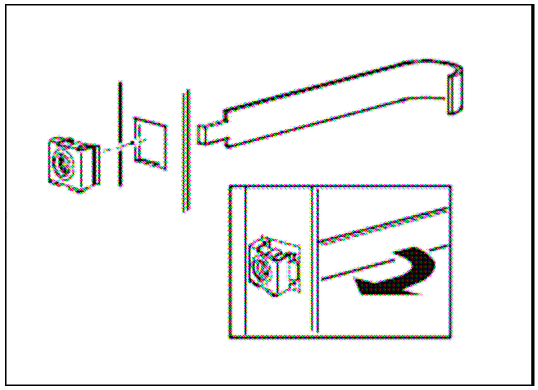


Figure 15: Using the Caged Nut Tool

To remove a caged nut, squeeze the sides and pull it out of the square hole.

The PDUs are fastened to the rack with M6 black anodized screws and washers.

Installing PDUs in a 42U APC rack

The PDUs in a 42U APC rack do not require mounting hardware. They come with circular mounting pegs that slide into tapered slots at the back of the rack:

Figure 16: 42U APC PDU - One PDU Installed

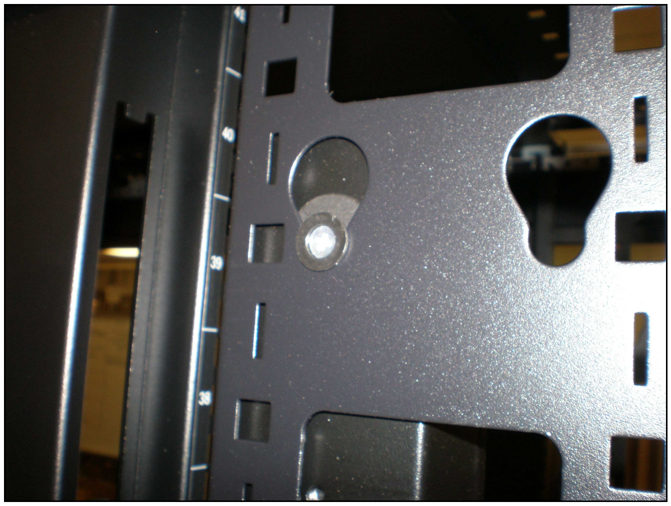


Figure 17: 42U APC PDU Peg and Slot - One PDU Installed

Figure 18: 42U APC PDU - Both PDUs Installed

Installing the Blade Rails

The procedure for installing the blade rails is the same for the 24U and 42U APC racks, except that with the 24U APC rack the first two U-spaces are occupied by the PDUs.

The blade rails are secured with threaded nut plates and 10-32 screws with washers:

Figure 19: Nut Plates and Screws for Blade Rails

1. To install the first blade rail in a 24U APC rack, insert three 10-32 screws into the three mounting holes in U-space 3 and screw them into the threaded holes in the nut plate. Each rail requires a nut plate with screws at the front and back of the rack. Leave the screws loose initially to accommodate the tabs on the blade rails.

Figure 20: Installing the Nut Plate and Screws

2. Insert the blade rail as shown. The tabs on the ends of the blade rails fit into the spaces between the nutplate screws. The end of the blade rail with the extender should be inserted at the back of the rack:

Figure 21: Blade Rail at Front of Rack

(screws loose)

Figure 22: Blade Rail at Back of Rack

(screws loose)

3. Tighten the 10-32 screws until the nut plate is clamped firmly around the blade rail tab.

   The side of the rail should be flush against the vertical mounting rail, and the top of the blade rail should be roughly even with – or just slightly below – the white line at the top of U-space:

Figure 23: Blade Rail Alignment

Figure 24: Right Blade Rail Installed – Front View

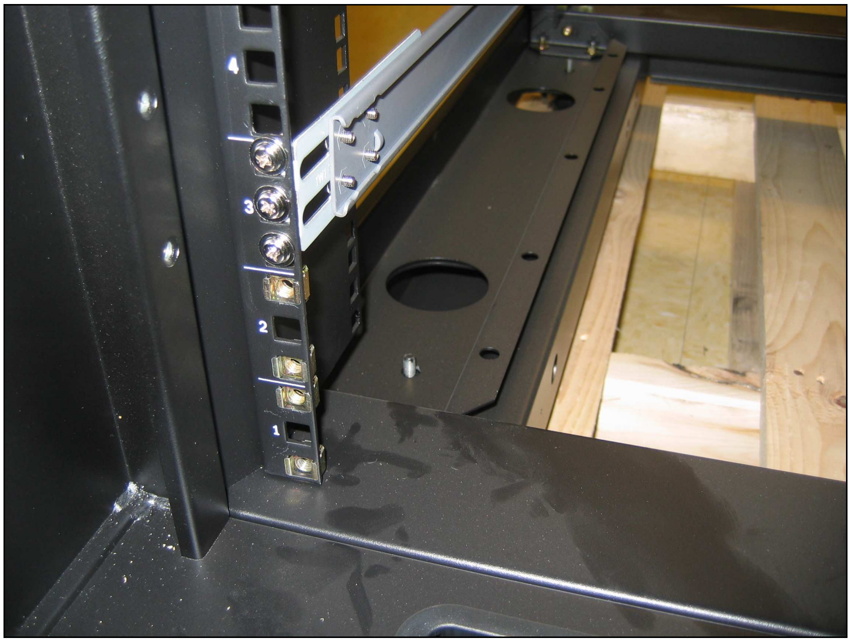


Figure 25: Right Blade Rail Installed – Rear View

4. Next, repeat the preceding steps for the left blade rail:

Figure 26: Left Blade Rail Installed – Front View

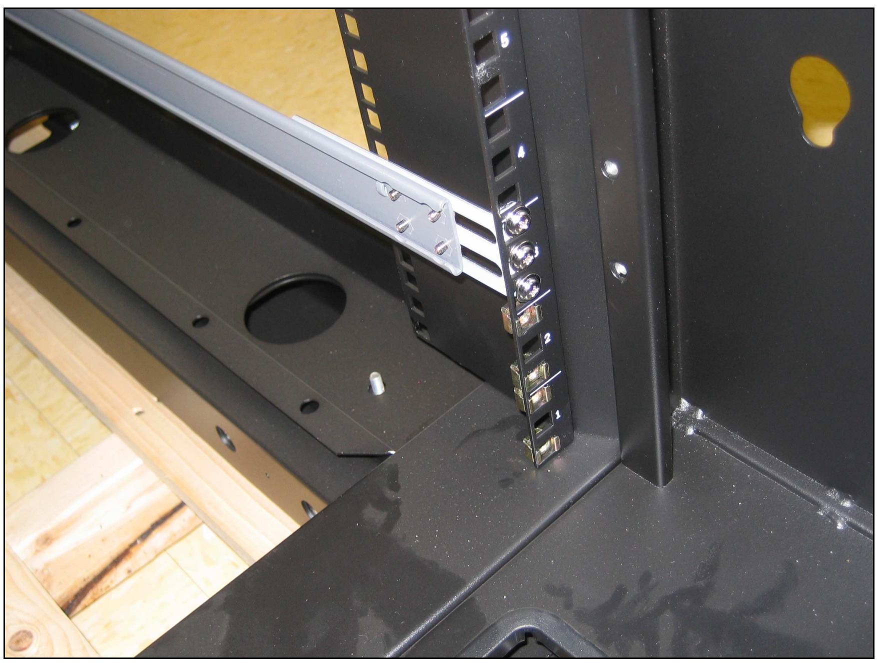


Figure 27: Left Blade Rail Installed – Rear View

You must skip a U-space to leave enough space for the next blade, so the next set of blade rails would be installed in U-space 5, and the set after that would be installed in U-space 7, and so on.

Figure 28: Multiple Blade Rails Installed in a 42U APC Rack

(rear view with KVM mounted in rack)

Installing the Blades

After installing the blade rails, slide the blade into the rack. The slides on each side of the blade fit into the inside of the rails on the rack. If you can't slide the blade into the rack, or if the blade is not level, you may need to remove the blade and readjust the blade rails.

Figure 29: Blade Installed in 24U APC Rack

(front view, PDUs not installed at bottom of rack)

Figure 30: Blades Installed in a 42U APC Rack

KVM

In both the 24U and 42U APC racks, the KVM is installed using a total of 12 caged nuts and M6 black anodized screws with washers – 3 caged nuts and 3 M6 black anodized screws with washers in a U-space in each of the four vertical rack mounting rails. The connectors on the KVM should face the back of the rack:

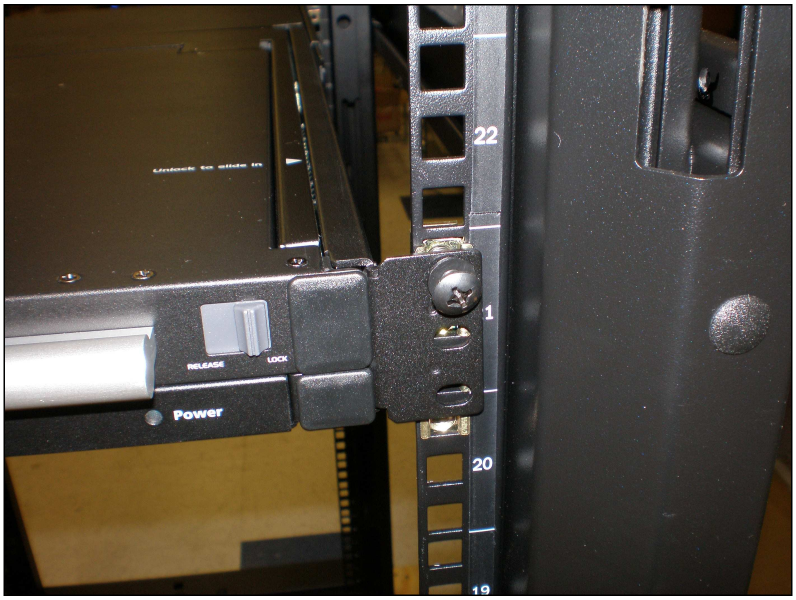


Figure 31: KVM Mounted in Rack with One Black M6 Screw & Washer

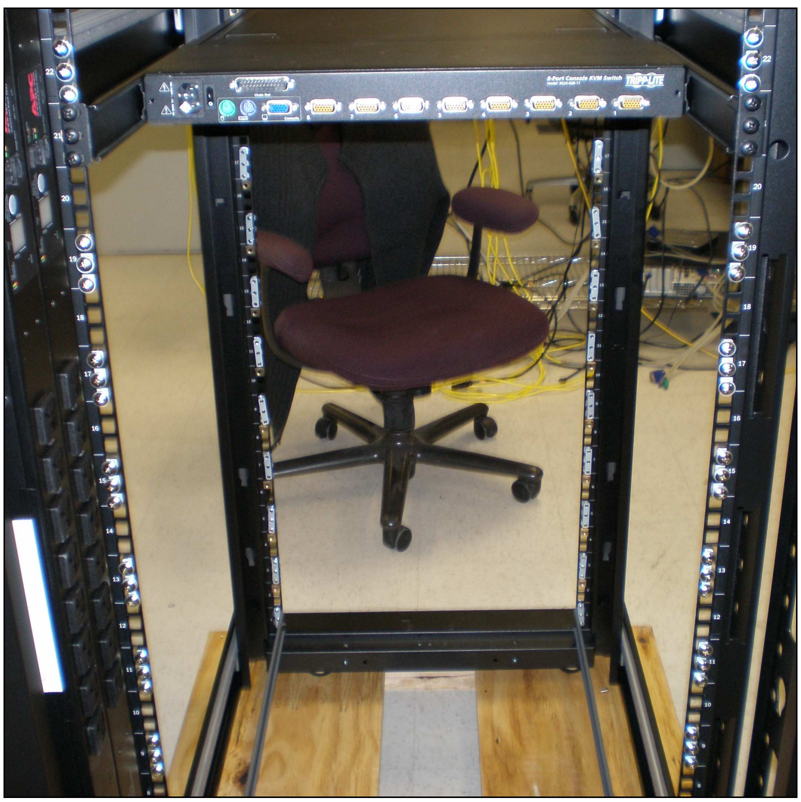


Figure 32: KVM Mounted in Rack with All Screws & Washers Installed

Network Switch

In both the 24U and 42U APC racks, the Network Switch is installed at the top of the rack using a total of 8 caged nuts and M6 black anodized screws with washers – 2 caged nuts and 2 M6 black anodized screws with washers in the top and bottom holes of a U-space in each of the four vertical rack mounting rails. The connectors on the Network Switch should face the back of the rack.

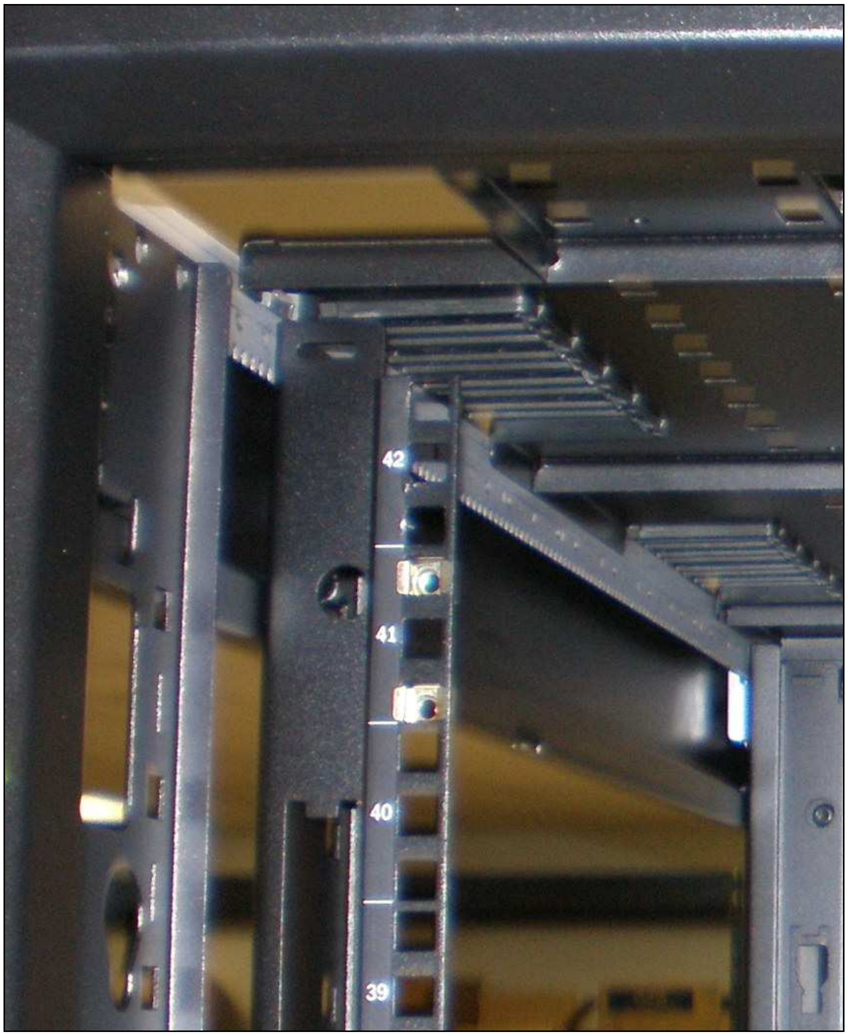


Figure 33: Caged Nuts for Network Switch in a 42U APC Rack

Figure 34: Network Switch Installed in a 42U APC Rack (rear view)

Terminal Server

In both the 24U and 42U APC racks, the Terminal Server is installed at the top of the rack just under the Network Switch using a total of 8 caged nuts and M6 black anodized screws with washers – 2 caged nuts and 2 M6 black anodized screws with washers in the top and bottom holes of a U-space in each of the four vertical rack mounting rails. The connectors on the Terminal Server should face the back of the rack.

Figure 35: Caged Nuts for Terminal Server in a 42U APC Rack

Figure 36: Terminal Server Installed Below Network Switch in a 42U APC Rack (rear view)

Loader Server and Dynamic Aggregation (DA) Server

The Loader Server and DA Server are optional components that are installed as extra server blades in the rack. These components are installed as previously described in the blade installation procedures.

Cable Routing

Dataupia currently ships systems that have been preassembled into either a 24U or 42U APC NetShelter SX racks. The systems ship with all of the cabling installed in the rack. This section describes cable routing for a Dataupia system in a 42RU APC Rack. For more details on interconnection between system components, refer to the Dataupia wiring diagrams that shipped with your system.

If you are using a rack other than an APC rack, the configuration should be similar to that of the APC rack, but you may need to modify the standard Dataupia cables to work with the alternate rack system, or provide your own cables. For more details on interconnection between system components, please refer to the Networking section of this document.

Note: cable colors can vary from system to system depending on customer requirements, so the cable colors shown in this section may not match those in the Networking section.

Power Cables

Power cables are routed from the PDUs to each system component along the left rear of the rack:

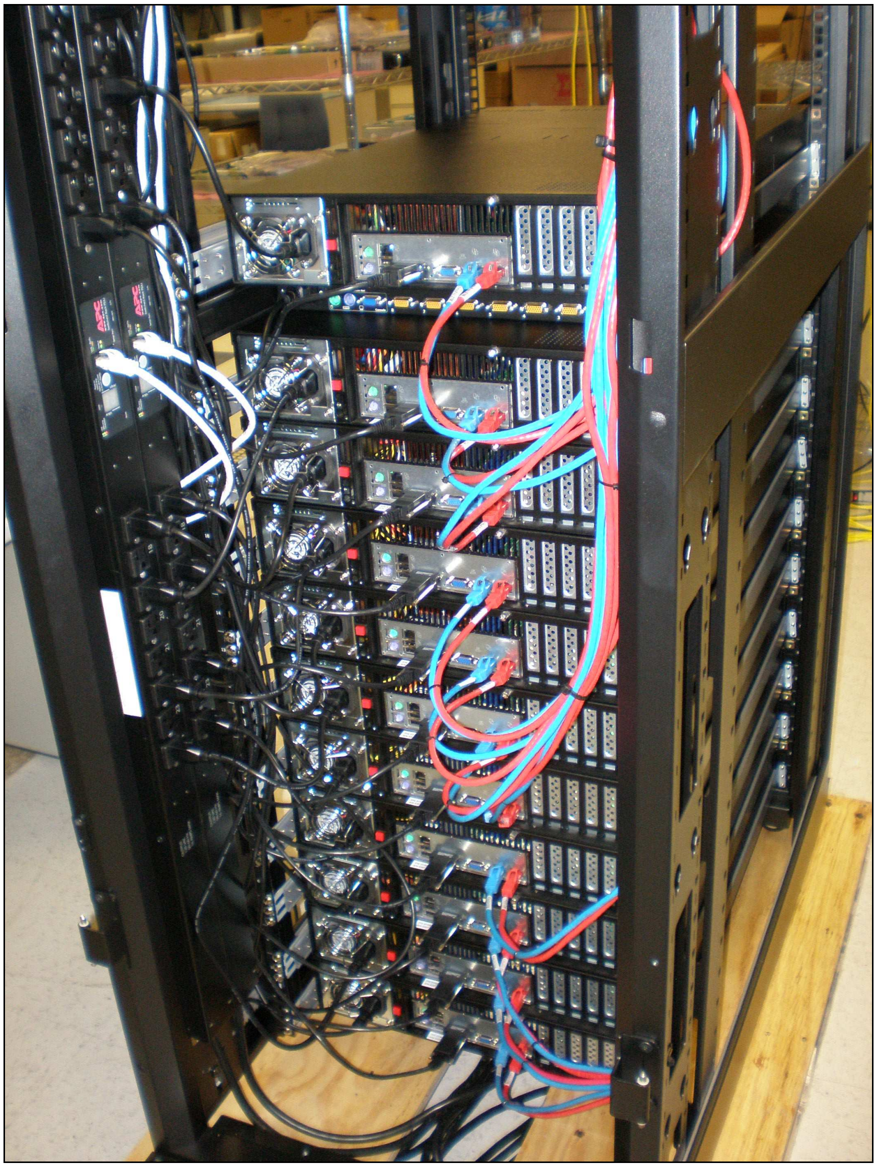


Figure 37: Power Cable Routing in a 42U APC Rack

Network Switch Cables

The cables from the Network Switch pass through a Panduit cable trough at the right rear of the rack and are routed to the eth1 and eth2 ports on each blade.

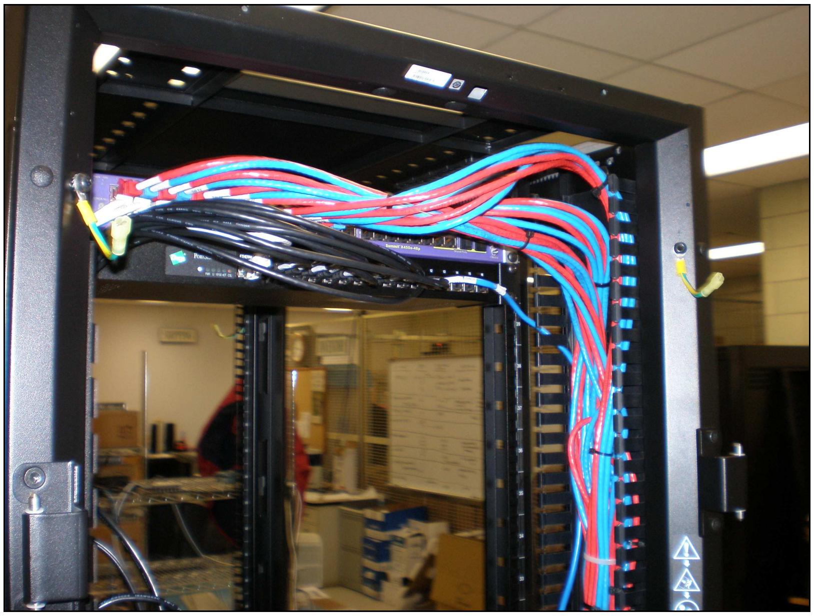


Figure 38: Network Switch Cables at Top of 42RU Rack

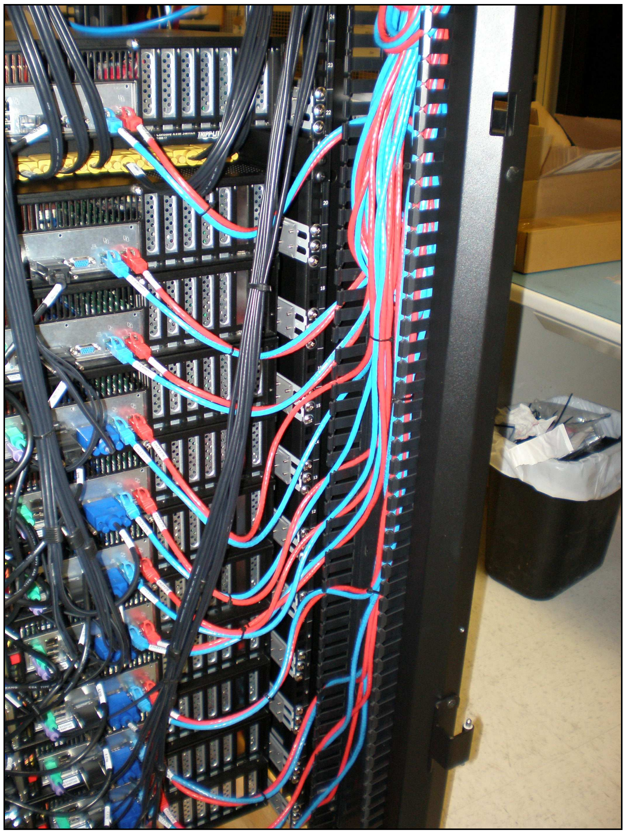


Figure 39: Network Switch Cables Connected to Blades

After routing the Network Switch cables, a cover is installed on the Panduit cable trough:

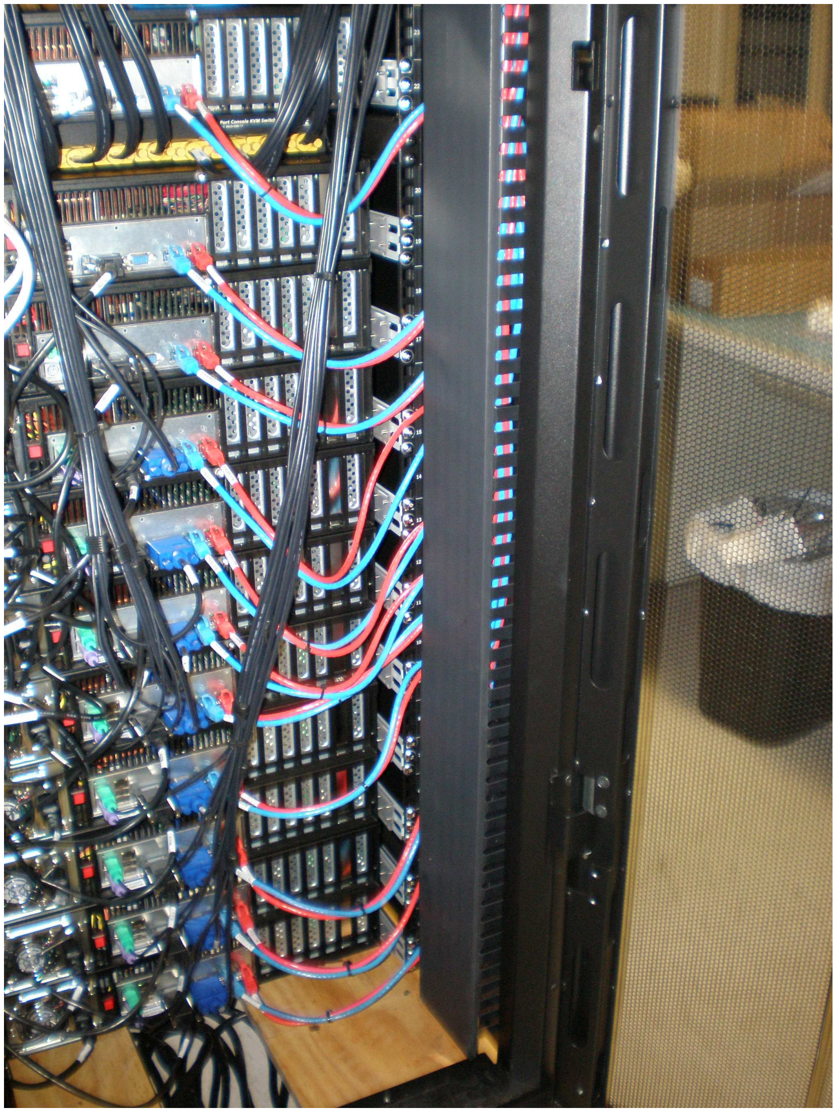


Figure 40: Panduit Cable Trough Cover

Terminal Server Cables

The cables from the Terminal Server pass through the channel next to the PDUs at the left rear of the rack and are routed to the terminal server ports on each blade.

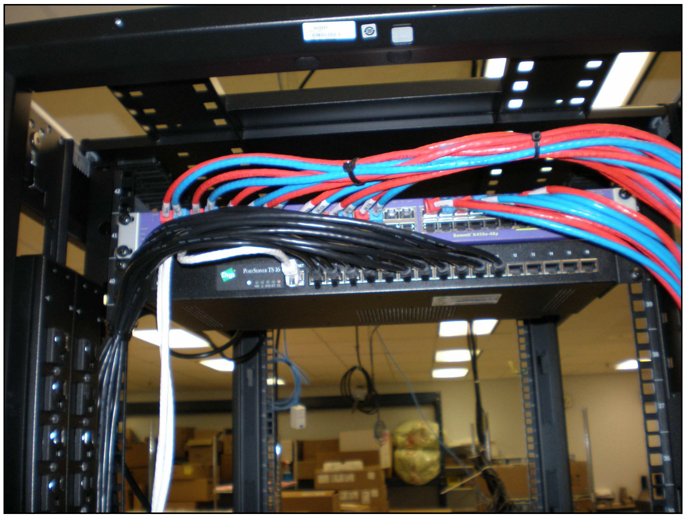


Figure 41: Terminal Server Cables at Top of 42RU Rack

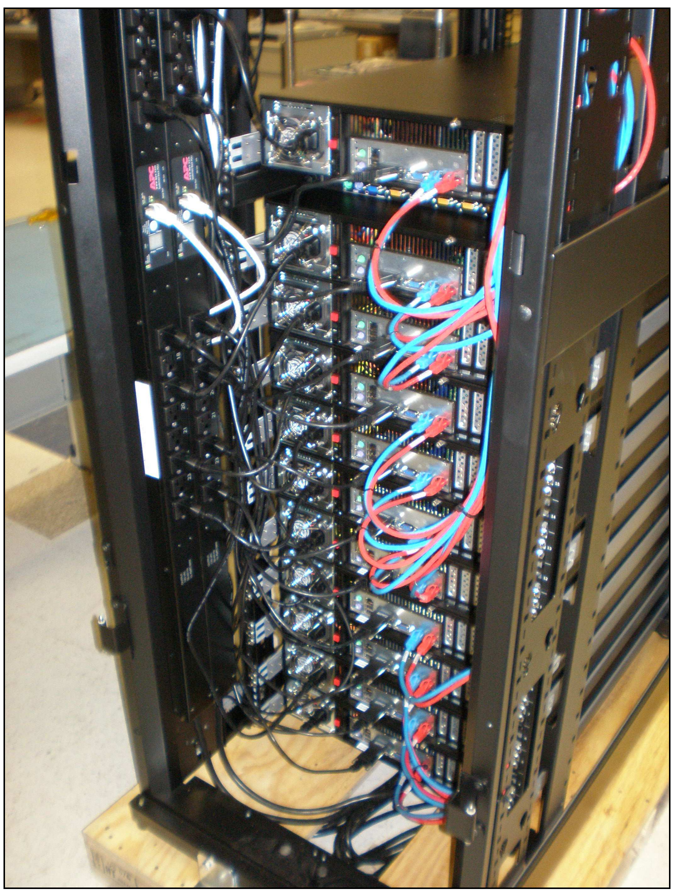


Figure 42: Terminal Server Cables Connected to Blades

KVM Cables

The cables from the KVM are routed directly to each blade. Excess cable lengths are looped and secured with cable ties above the KVM at the sides of the rack.

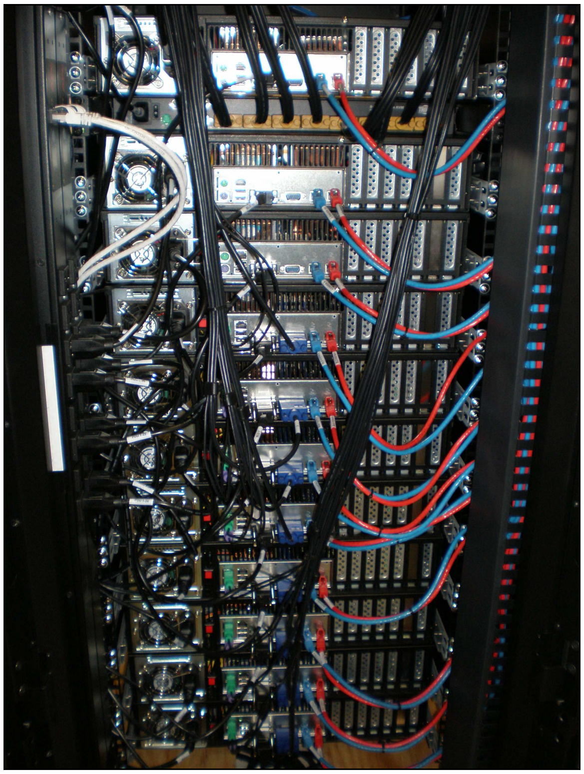


Figure 43: KVM Cables Routed to Blades

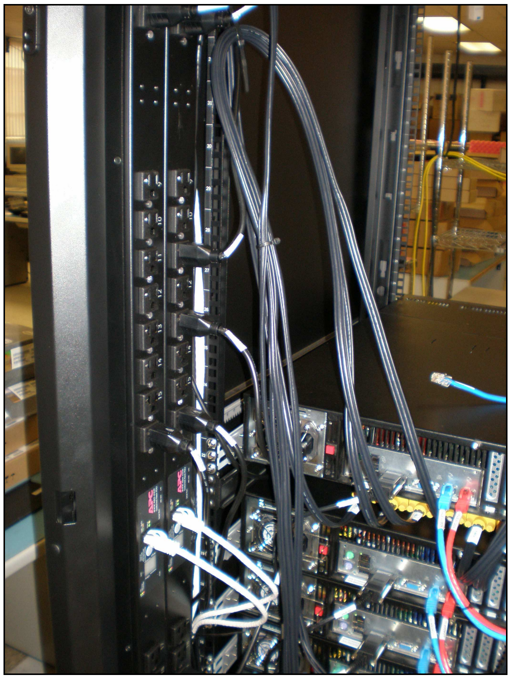


Figure 44: Excess KVM Cable

Loader Server and DA Server Cables

The Loader Server and DA Server are optional components that are installed as extra server blades in the rack. These components are installed as described in the Networking section of this document.

Appendix B: NFS Mount Set Up and Configuration

The following are the steps needed to configure the Loader blade or the DA blade as an NFS Server to which you can connect from other Linux systems or Solaris systems.

Login to the Loader Blade (or DA blade) and become root.

su - (including the space and dash)

Confirm you have the following packages installed with the same (or newer) versions. (This should be the manufacturing default for new Loader and DA blades.)

```
[root@loader ~]# rpm -qa | egrep nfs
nfs-utils-lib-1.0.8-7.2
nfs-utils-1.0.10-14.fc6

[root@loader ~]# rpm -qa | egrep kernel
kernel-headers-2.6.22.14-72.fc6
kernel-devel-2.6.22.14-72.fc6
kernel-devel-2.6.18-1.2798.fc6
kernel-2.6.22.14-72.fc6
kernel-2.6.18-1.2798.fc6
```

Create the file /etc/exports; this file is where you configure the directory you want to expose and also who (hosts/IPs) will have access to connect to it.

```
# vi /etc/exports
```

Into the file, add the following line(s) to specify the directory to expose, who has access to mount it, and the type of access, e.g., read, write, etc. Take care not to add whitespace in the file as it is sensitive.

Format `{/directory/path} {system_IP}(option1,option2...)`

Example 1: `/mnt/data/stuff 10.0.2.10(ro)`

Example 2: `/mnt/data/stuff 10.0.0.0/255.255.0.0(rw)`

In Example1, the directory `/mnt/data/stuff` is exposed and the single system (IP) of 10.0.2.10 can mount the share. The `(ro)` indicates read only access to the share.

In Example2, the directory `/mnt/data/stuff` is exposed and the entire IP range 10.0.x.x/255.255.0.0 has access to mount the share. The `(rw)` indicates read/write access to the share.

Some additional options available are:

- **ro** The directory is shared read only; the client machine will not be able to write it. **This is the default**.
- **rw** The client machine will have read and write access to the directory.
- **no_root_squash** By default, any file request made by user root on the client machine is treated as if it is made by user nobody on the server. (Exactly which UID the request is mapped to depends on the UID of user "nobody" on the server, not the client.) If **no_root_squash** is selected, then root on the client machine will have the same level of access to the files on the system as root on the server. This can have serious security implications, although it may be necessary if you want to perform any administrative work on the client machine that involves the exported directories. **You should not specify this option without good reason**.
- **no_subtree_check** If only part of a volume is exported, a routine called subtree checking verifies that a file that is requested from the client is in the appropriate part of the volume. If the entire volume is exported, disabling this check will speed up transfers.

Making Changes to `/etc/exports` later on: If you come back and change your /etc/exports file, the changes you make may not take effect immediately. You must run the command `exportfs -ra` to force NFSD to re-read the `/etc/exports` file.

Once the /etc/exports file is created and the share has been added, confirm the directory on the Loader blade has the correct permissions. If you only want read access, the default settings are correct. If you want to have read/write access from the client onto the NFS Server (the loader), you will need to change the permissions on the directory you will be exposing and mounting to.

To follow with our example, if the directory you want to expose on the NFS Server (the Loader) with read/write access is `/mnt/data/stuff` then you will need to change the permissions on the `/stuff` directory to allow read/write access to the nfs mount.

```
# chmod 777 stuff/
```

Starting the NFS Server Service

Next you need to start the NFS server.

```
[root@loader ~]# /etc/init.d/nfs start
Starting NFS services:  [  OK  ]
Starting NFS quotas: [  OK  ]
Starting NFS daemon: [  OK  ]
Starting NFS mountd: [  OK  ]
[root@loader ~]#
```

You also have **/etc/init.d/nfs status** and **/etc/init.d/nfs stop** available to you to check the status and stop the service if needed.

Confirm the NFS Server has started. You will see a number of nfsd processes started. The Process ID numbers you see will be different on your system.

```
[root@loader ~]# ps -ef | egrep nfsd
root      3436     2  0 16:23 ?        00:00:00 [nfsd4]
root      3437     2  0 16:23 ?        00:00:00 [nfsd]
root      3438     2  0 16:23 ?        00:00:00 [nfsd]
root      3439     2  0 16:23 ?        00:00:00 [nfsd]
root      3440     2  0 16:23 ?        00:00:00 [nfsd]
root      3441     2  0 16:23 ?        00:00:00 [nfsd]
root      3442     2  0 16:23 ?        00:00:00 [nfsd]
root      3443     2  0 16:23 ?        00:00:00 [nfsd]
root      3444     2  0 16:23 ?        00:00:00 [nfsd]
root      3483  3302  0 16:29 pts/0    00:00:00 egrep nfsd
[root@loader ~]#
```

Use the `showmount` command to show the mount points available on an NFS Server. It can be run local on the NFS Server to show what it is exposing and also from a client (another Linux or Solaris System) to see what a particular NFS Server is exposing.

On the NFS Server (Loader/DA)

```
[root@loader ~]# showmount -e
Export list for loader:
/mnt/data/stuff 10.0.0.0/255.255.0.0
```

On the Solaris system (client) *pointing* to the NFS Server (Loader/DA)

```
bash-3.00# showmount -e 10.0.2.20...(the IP of the NFS Server)
export list for 10.0.2.20:
/mnt/data 10.0.0.0/255.255.0.0
bash-3.00#
```

You can also use the `rpcinfo` command to confirm the NFSD daemon (process) is running. This can also be run from the local NFS Server and from a client (another Linux or Solaris System) to confirm NFS connectivity between systems.

On the NFS Server (Loader/DA)

```
[root@loader ~]# rpcinfo -p
   program vers proto   port
    100000    2   tcp    111  portmapper
    100000    2   udp    111  portmapper
    100024    1   udp    761  status
    100024    1   tcp    764  status
    100011    1   udp    637  rquotad
    100011    2   udp    637  rquotad
    100011    1   tcp    640  rquotad
    100011    2   tcp    640  rquotad
    100003    2   udp   2049  nfs
    100003    3   udp   2049  nfs
    100003    4   udp   2049  nfs
    100021    1   udp  32770  nlockmgr
    100021    3   udp  32770  nlockmgr
    100021    4   udp  32770  nlockmgr
    100003    2   tcp   2049  nfs
    100003    3   tcp   2049  nfs
    100003    4   tcp   2049  nfs
    100021    1   tcp  55907  nlockmgr
    100021    3   tcp  55907  nlockmgr
    100021    4   tcp  55907  nlockmgr
    100005    1   udp    656  mountd
    100005    1   tcp    659  mountd
    100005    2   udp    656  mountd
    100005    2   tcp    659  mountd
    100005    3   udp    656  mountd
    100005    3   tcp    659  mountd
[root@loader ~]#
```

On the NFS Server (Loader/DA), confirm you see an entry, with version 3 (version 2 for portmapper), for each of the following: `portmapper`, `nfs`, `rquotad`, and `mountd`.

On the Solaris system pointing to the NFS Server (Loader/DA); this can be done from any client that can *see* the NFS Server.

```
bash-3.00# rpcinfo -p 10.0.2.20  ..IP address of NFS Server (Loader)
   program vers proto   port  service
    100000    2   tcp    111  rpcbind
    100000    2   udp    111  rpcbind
    100024    1   udp    761  status
    100024    1   tcp    764  status
    100011    1   udp    637  rquotad
    100011    2   udp    637  rquotad
    100011    1   tcp    640  rquotad
    100011    2   tcp    640  rquotad
    100003    2   udp   2049  nfs
    100003    3   udp   2049  nfs
    100003    4   udp   2049  nfs
    100021    1   udp  32770  nlockmgr
    100021    3   udp  32770  nlockmgr
    100021    4   udp  32770  nlockmgr
    100003    2   tcp   2049  nfs
    100003    3   tcp   2049  nfs
    100003    4   tcp   2049  nfs
    100021    1   tcp  55907  nlockmgr
    100021    3   tcp  55907  nlockmgr
    100021    4   tcp  55907  nlockmgr
    100005    1   udp    656  mountd
    100005    1   tcp    659  mountd
    100005    2   udp    656  mountd
    100005    2   tcp    659  mountd
    100005    3   udp    656  mountd
    100005    3   tcp    659  mountd
bash-3.00#
```

From the client, confirm you see an entry, with version 3, for each of the following: `nfs`, `mountd`, and `rquotad`.

Configuring NFS to start on system reboot

To have the NFS Server daemon start when the Loader is booted, you need to enable the NFS service for runlevels 3 & 5.

To check the current runlevel for NFS:

```
[root@loader ~]# chkconfig --list nfs
```

Output:

```
nfs      0:off   1:off   2:off   3:on    4:off   5:on    6:off
```

To enable nfs for runlevels 3 & 5:

```
[root@loader ~]# chkconfig --level 35 nfs on
```

Confirm the change happened by checking again; `chkconfig --list nfs`. The NFS services will start on reboot of the system and automatically expose/share the directories defined in /etc/exports.

Connecting from a Linux or Solaris client

To connect to an NFS Server (Loader Blade) share from a client, use the mount command. Below are the commands to connect and disconnect (mount and umount) from both a Linux and Solaris system.

> ***Note:*** Be sure that both the exposed (shared) directory exists on the NFS Server (and has the correct permissions; see above) and you have created an empty directory to use locally as a mount point on the client. Be aware, the local directory should be a newly created directory for the purpose of connecting to the remote share. Once you have mounted the remote share over the local directory, you will no longer have visibility into the original/local directory.

From a **Linux** system

Format `mount {IP}:{/remote/share} {/local/mount/point}`

Example `mount 10.0.2.20:/mnt/data/stuff /mnt/data/test/`

To un-mount the share use the `umount` command: `umount /mnt/test/`

From a **Solaris** system

Format

`mount -o {`*`options`*`} {IP}:{`*`/remote/share`*`} {`*`/local/mount/point`*`}`

Example `mount -o vers=3 10.0.2.20:/mnt/data/stuff /mnt/test`

To un-mount the share use the `umount` command: `umount /mnt/test/`

Appendix C: Setting IP addresses

Loader or DA Server (FC6/Red Hat)

Find the configuration file corresponding to the interface you need to change:

/etc/sysconfig/network-scripts/ifcfg-eth<x>

Important: If the file contains a HWADDR= or MACADDR= statement, do not modify or remove these.

Assuming the interface is configured as static IP; modify these statements to change the IP address:

```
IPADDR=<new IP address>
NETMASK=<new netmask>
```

Sample ifcfg-eth<x> file for static IP:

```
DEVICE=eth0
BOOTPROTO=none
ONBOOT=yes
NETMASK=255.255.255.0
IPADDR=10.0.1.27
USERCTL=no
```

Sample ifcfg-eth<x> for DHCP:

```
DEVICE=eth0
BOOTPROTO=dhcp
ONBOOT=yes
```

To modify default gateway:

Note: If DHCP is used, DHCP will assign the gateway address.

/etc/sysconfig/network

```
GATEWAY=<default-gateway-ip-address>
```

SUN/SPARC Server

There are four files that need to be updated on a Solaris system to change the IP address. All changes MUST be done as root.

su -   (including dash, provide password to become root)

ifconfig -a   before changing anything run this command and note the IP configuration including the netmasks.

`/etc/hostname.`*`{eth_name}`*

This file contains the hostname associated to the NIC for this system. The IP address set at the NIC will be the address that the hostname resolves to (usually in /etc/hosts). If you are just changing the IP of the NIC change it in /etc/hosts.

`/etc/hosts and /etc/inet/hosts`

This file associates an IP address with a host name. The 10.0.2.x IP address is the connection to the array; do not change this IP address. The other IP address (10.214.x.x) is the IP address you need to change to the IP address the customer wants the DB host system to have. This is a customer provided IP address that is in the customer provided IP range we're using for the CUSTOMER VLAN.

`/etc/netmasks`

This file defines the subnet mask for each IP address specified. The 10.0.0.0 IP address with the 255.255.0.0 subnet mask is the connection to the array; do not change this IP address or subnet mask.

The other IP address (10.214.128.0) and subnet mask (255.255.252.0) are the addresses you need to change to the addresses the customer specifies.

To determine the IP address to use, take the customer provided subnet mask (enter this as the customer provides it) and where there are 0's in the mask, specify 0's in the network address.

For example: given the following IP address 10.214.134.20 and the subnet mask of 255.255.0.0. The entry in the file would be 10.214.0.0 255.255.0.0.

`/etc/defaultrouter`

This file contains the default gateway for the customer network (also used for the CUSTOMER VLAN). This is a customer provided IP address that is in the customer provided IP range we're using for the CUSTOMER VLAN.

When all necessary changes are made, restart the network services:

```
# svcadm restart network/physical
```

Satori Server blade

WARNING: Do not attempt to change the IP address on a Satori server which has been joined to a Satori server array - contact Support for a special procedure to change the IP address on all Satori servers in the array simultaneously. Do not attempt to use the initial configuration procedures.

To initially set the IP on a Satori server blade that is in the factory default state, see the *Dataupia Satori Server Installation and Configuration Guide*.

Appendix D: Replacing a Failed Disk Drive

Each blade in a Dataupia Satori System contains eight RAID 5 drives. There are seven active drives and one hot spare. The drive bays in the top row of each blade are numbered four through seven from left to right. The drive bays in the bottom row are numbered zero through three from left to right.

If a drive fails, the blade continues to operate in degraded mode while the RAID is rebuilt on the spare drive (which may take many hours). Replace the failed drive as soon as possible.

> ***Note:*** To ensure data integrity, the drive should not be replaced until the rebuild process is complete.

To replace a drive, remove the failed drive and replace it with a new drive from the factory. It is not necessary to log in to the system or to issue any commands to replace a bad drive, but you must know which drive it is. This can be determined from the failed drive email or with the assistance of Dataupia Support.

To remove a failed drive, push the release button at the front of the drive to the right and lift the release lever. The release lever will swing out and to the left. After the release lever is fully extended, pull the disk drive out of the drive bay.

Remove the drive label from the failed drive and place it on the new drive. Place the failed drive into an anti-static container and return it to the factory.

To install a new drive, push the new drive as far as possible into the drive bay, then swing the release lever all the way to the right to lock the drive in place. The release lever will be flat up against the front of the blade chassis.

Once the drive has been replaced, if email alerts are configured, you should receive an email alert indicating that the drive or RAID error has cleared. If you don’t receive this notice, please contact Dataupia Support. (You can use the Dataupia Management Console or the Command Line Interface to set up email notifications of drive failure or repair alerts.)

Appendix E: Replacing a Failed Blade Chassis

About Blade Chassis Failure

Each blade in a Dataupia Satori System contains eight RAID 5 drives. There are seven active drives and one hot spare. The drive bays in the top row of each blade are numbered four through seven from left to right. The drive bays in the bottom row are numbered zero through three from left to right.

In the original configuration at shipment, the label on each disk drive matches the drive bay number. Drive bay seven contains the original hot spare. The label on each disk drive matches the drive bay number.

If a blade fails and diagnostic tests have eliminated a failed drive or a network issue as the cause of the problem, the blade chassis must be replaced.

Generally, a chassis must be replaced if the CPU, Host Bus Adapter, memory, disk controller, power supply, or cooling fan fails.

No data is lost during a chassis replacement, but the array is inaccessible while the chassis is being replaced.

If a blade is nonfunctional and cannot be cleanly shut down, a file system check, a RAID verification, or both may be required when you boot the swapped blade. These procedures can take considerable time, perhaps hours. Do not under any circumstances interrupt these checks; doing so could jeopardize the data on the disks.

The new chassis may have a software image version that is different from the image on the failed blade and the rest of the array. It's not possible to know until you install and boot the new blade. If the image on the chassis is earlier than the one on the array, you'll need to upgrade it to an image that matches the other blades before restarting the array. If the image on the chassis is later than the image on the array, you'll need to upgrade the other blades once you restart the array.

To Replace a Blade Chassis

1. Shutdown and power-off the blade

1. Use the **Upgrade** option on the **Summary** tab of the Dataupia Management Console or the **show version** command in the Command Line Interface to determine which software image is booted on the array. Make a note of the software image. All blades must be running the same image.
2. Use either the Dataupia Management Console (DMC) or the Command Line Interface (CLI) to shutdown the blade.
    - To shutdown using the DMC, select the blade in the navigation pane, then select **Shut Down** from the drop-down menu on the **Summary** tab. Be sure the blade is selected before selecting **Shut Down** so you don't shut down the entire array.
    - To shutdown using the CLI, use the **blade shutdown** command. The blade shutdown command shuts down and powers off the blade.

   If you cannot shut down the blade cleanly with either the DMC or the CLI, you can power-off the blade by pressing and holding the power button at the front of the blade until the blue power indicator lights on the front of the blade shut off. It may take a few seconds for the blade to power-off.

2. Disconnect the cables at the back of the blade chassis

Unplug all of the cables at the back of the applicable blade chassis. Note the location of the cables so you can reconnect them the same way.

- To remove the eth1 and eth2 Ethernet connections, squeeze the strain relief clip at the top of each connector and pull the connector out.
- Unscrew the retaining screws and remove the terminal server connector.
- Pull out the AC power connector.

3. Remove the blade chassis from the rack

1. Remove any rack mounting screws from the front sides of the chassis.
2. Slide the chassis out. The chassis will stop when it reaches the retaining clips on each side of the slide.
3. Push down to release the left retaining clip, push up to release the right retaining clip, then slide the chassis out the rest of the way.

   Note: blade chassis weighs approximately 50 pounds. It might be easier for two people to remove the chassis.

4. Move the disk drives from the failed blade chassis into the new chassis

1. One by one, remove the disk drives from the failed chassis and install them in the new chassis, making sure they are in the exact same order and slot as the the failed chassis. The drive bays in the top row are numbered four through seven from left to right. The drive bays in the bottom row are numbered zero through three from left to right.
2. To remove a disk drive from the failed chassis, push the release button to the right and lift the drive release lever. The release lever will swing out and to the left. After the release lever is fully extended, pull the disk drive out of the drive bay.
3. To install a drive into the new blade chassis, push the new drive as far as possible into the drive bay, then swing the release lever all the way to the right to lock the drive in place. The release lever will be flat up against the front of the blade.
4. Make sure that the drive number on each disk drive matches the drive bay number.

5. Install the new blade chassis in the rack

1. Slide the new chassis into the front of the rack.

   Note: the A Satori Server chassis weighs approximately 50 pounds. It might be easier for two people to remove the chassis.

6. Connect the cables to the back of the new blade chassis

Reconnect all of the cables at the back of the new chassis.

- Plug in the eth1 and eth2 Ethernet connections until the strain relief clips snap into place. The green LEDs will begin to flash when eth1 and eth2 are connected.
- Plug in the AC power plug and the terminal server connector. Tighten the retaining screws on the terminal server connector.

7. Power up the blade

After reinstalling all of the disk drives, press the power button to power up the blade.

If the blade was not shut down cleanly and a file system check or RAID verification needs to run, the boot process will halt after the kernel boots and you will see a lot of activity on the disk drives.

Caution: Do not interrupt this process or data integrity may be compromised.

8. About software image versions

Each blade has two partitions on which software image versions are installed. The initial images are installed during manufacturing. To upgrade a blade Server, you install a new image on the partition that is not the current boot partition, change boot partitions, and then reboot the blade from the new partition.

All blades in an array **MUST** be running the same software image version at all times. Blades cannot be downgraded, and images are not backward compatible.

If the image on the new chassis is earlier than the image on the array, you'll need to upgrade the new blade to an image that matches the other blades in the array, then restart the the new blade to activate the new image. If the image on the chassis is later than the image on the array, you'll need to upgrade all of the other blades, then restart each blade that was upgraded.

Use the **Summary** tab on the DMC or the **show version** command in the CLI to make sure that all servers in the array are running the same image.

If the images don't match, you can upgrade the array or a single server by selecting **Upgrade** from the menu on the **Summary** tab in the DMC. You can upgrade the array or an individual blade, depending on which summary is displayed.

Once you have upgraded so that all of the blades are running the same image, you can use the **reboot** link at the bottom of the page to reboot either the array or the blade to the new image.

Appendix F: Troubleshooting

Each blade in a Dataupia array contains eight RAID-5 disk drives—seven active drives and one hot spare. If you are experiencing operational indications of a problem with a Dataupia array (such as poor performance, incomplete query results, or inconsistent behavior), one of the active drives may have failed. This may also be indicated by a **No drive available for SPARE creation on controller** incident or **Disk space low** incident on a blade in the Dataupia Management Console (DMC), as described in Chapter 3.

Because the blade contains RAID-5 drives with a hot spare, the failure of one drive does not result in data loss, because redundant data is stored on the remaining active disks and the failed disk is rebuilt on the hot spare. However, the blade runs in degraded mode, with no spare (thus the **No drive available for SPARE** incident), until the spare is fully rebuilt to replace the failed drive and a new spare is available. Failed drives should therefore be replaced as quickly as possible.

When one of the disk drives in a blade fails and is rebuilt on the existing spare, a new drive must be installed to replace the spare.

You will receive an email notification (and a message on the DMC) when a drive has failed. In the Detail section of the email it will indicate which port number has failed. Here is s sample email:

```
==== System information:
Hostname: alewife
Version:  dataupia 2.0.7 #11812 2008-02-11 22:46:51
          root@dtbuild01:Release
Date:     2008/02/14 08:03:50
Uptime:   1d 13h 39m 7.632s
Description: Raid array controller event
Severity: Error (1)
Detail: Degraded unit: unit=0, port=4 (2)
Timestamp: 2008/02/14 08:03:39
Node: /dataupia/raid/events/error
```

The email shows that the drive in port #4 has failed. This number corresponds to the drive number marked on the front of the blade.

See Appendix D for instructions of replacing a failed drive.

Repairing a Network Connection

If you have operational indications of a problem and it doesn't appear to be a failed disk drive, or there is a **No network response from blade** incident on a blade, the cause might be a problem with a network connection rather than with a blade. Take the following steps to determine whether this is the case:

1. On a computer connected to a network within the appropriate domain, issue a **ping** command to the IP address of the indicated blade, or to all blades if you have not yet isolated a particular blade as the problem. If you can reach a blade, the connection is functioning and there is no need to go any further with this procedure.
2. Physically check the blade to confirm that it is powered on and operating.
3. Connect a monitor and keyboard to the appropriate ports in the back of the blade.
4. Log in as **dtadmin**.
5. Issue a **ping** command to another IP address.

6. If you cannot reach the network from the blade, try a **blade restart** command to restart services or a **blade reload** command for a warm reboot of the blade.
7. If the network connection is still down, check the blade's cable connection to the switch between the blade and the network. Are the lights green at both ends? If not, try moving the cable to a different switch port and replacing the cable.
8. If the network connection is still down, check the health and status of the switch. Try bypassing the switch by disconnecting the network connection from the switch and connecting it directly to the blade. If this does not restore the blade's network connection, the blade's host bus adapter (HBA) has failed and the blade must be replaced (see the Replacing a Blade section).

Restarting a Blade

If you have been unable to correct a problem with a blade and the network connection is functioning, you can try a warm restart of the blade using the DMC (Chapter 3) or the **blade reload** command.

Replacing a Blade

If you have determined that a problem with a blade is not caused by a failed drive or network connection, and restarting the blade has not solved the problem, you must replace the blade.

Generally a blade must be replaced if any of the following components fail:

- CPU
- HBA
- memory
- disk controller
- fan
- power supply

The following indications on a blade, therefore, usually call for blade replacement:

- Fan 1 is not on.
- Fan 2 is not on.
- CPU temperature is approaching a critical level.
- No 9xxx controller found.
- No RAID-5 units found on controller

See Appendix E for steps to replace a failed chassis. If you need to replace a blade chassis, contact Dataupia Technical Support:

Phone: 866-259-5971
Fax: 617-441-7776
Email: **support@dataupia.com**
Customer Portal: www.dataupia.com Click the **Customer Login** link at the top right of the page.

Evaluating Query Performance

Use the DMC to monitor and evaluate the speed with which queries against the data on your array are executed, and investigate and remedy problems when they occur.

The time needed for a query to complete depends on a number of factors, including the complexity of the query, the number, the size and indexing of the tables against which it runs, the number of results it returns, and the concurrent load on array resources. Factors external to the array, such as concurrent load on the network or linked database server, also play a role. For this reason, it is important to establish a query performance baseline for your array.

The **Queries** tab helps you establish this baseline and evaluate ongoing query performance. The grids on this tab provide duration and other basic information for both historical (completed) queries and active queries (those that have been executing for at least 15 seconds). Use this information to develop performance norms for various types of queries, to compare performance of recent queries to the norms you have established as well as previous queries, and to spot problem queries (those that require a long time to complete or become stalled altogether). In addition to slow or stalled queries, you may need to investigate queries that return errors, incorrect results, or extraneous results.

Analysis and Remediation

There are a number of possibilities to be considered in identifying the causes of unsatisfactory query performance and taking action to improve it. Table 8 below provides suggestions for investigating possible causes and applying remedies.

Figure 1 Query Performance Troubleshooting

Problem	Investigation and Remediation
Queries run concurrent with resource-intensive tasks, events, and conditions, including: • complex queries against large tables • multiple array logs configured at Info or Debug detail level, DDL/DML logging on • high network traffic	Review problem queries to identify particular periods or days associated with them, if any. Review logging configuration. Review logs to identify resource-intensive system events. Review database and network traffic patterns.
	Redistribute query load over time as indicated. Configure fewer logs, turn off DDL/DML logging. Manage and redistribute other resource-intensive system events.
Array or blade health problem	See other sections in this chapter for information about investigating and remedying array and blade health problems as indicated by statistics, alerts, or substandard query performance.
Inefficient query syntax or plan	Review syntax of problem queries and target tables associated with them. Review DDL/DML logging.
Inefficient table indexing	Review problem queries to identify particular tables associated with them.
	Review indexing of tables and remove or add indexes
Inefficient distribution of values within frequently queried columns	Review problem queries to identify frequently queried columns in target tables Query key columns and review distribution of values

Problem	**Investigation and Remediation**
	Unload data from tables and reorder records to cluster similar values in key columns Truncate tables and reload reordered data
Inefficient data distribution method for table	Review problem queries to identify particular target tables associated with them Review data distribution of indicated tables
	Unload data from tables and drop tables Recreate tables with alternate data distribution and reload data

Preliminary Query Testing

One of the best ways to avoid some of the problems listed in Table 8 is to create several test tables, load small amounts of data, and run the same query against them to compare execution time using the historical queries grid on the **Queries** tab. This can help you resolve potential problems with inefficient indexing, value distribution within columns, and data distribution before loading large amounts of data.

Appendix G: Alert email headers

Satori Server blades send alerts of critical parameters to an email list. Emails are sent when a condition is recognized and again when the condition clears (this is configurable on an alert-by-alert basis -- see *Dataupia Satori Server User Guide* for details on configuring alerts).

Satori Server alert email capability is divided into two parts: the email alerts for system administrators, and an auto-support email facility for maintainers. Each part can be enabled independently, and each part is disabled by default. System administrators should configure email alerts to point to their group so that they are informed of critical events in a timely fashion. Configuration of these email alerts can be changed by the system administrators to best reflect their needs. Autosupport email, on the other hand, reports diagnostic information intended for maintainers and factory-trained personnel. Your installer will work with you on how best to configure this facility and its configuration is not exposed. Dataupia would like to receive as much diagnostic information as the customer can permit, and for that purpose autosupport emails, when enabled, will relay to Dataupia Technical Support at dataupia.com. If it is not possible or permissible to relay emails to Dataupia, other arrangements can be made to make this information available to Dataupia Technical Support personnel.

To receive email alerts an SMTP server must be configured to the Satori Server blade. Also, with the wide variety of SMTP servers, and email pre-processing such as authentication and spam filtering and the different requirements of enterprise email systems, it may be necessary to configure the SMTP server to do the right thing with the emails sent by Satori Servers, or to override some of the Satori Server email defaults, or both. This section was written with the hope of making it easier to understand and configure Satori Server email alert capability.

Configuring the email header

Here is a copy of the default email header that the blades will construct for either administrator or auto-support emails:

```
Received: by qatek04 (sSMTP sendmail emulation); Tue,  7 Apr 2009 20:36:19
+0000
From: "Admin User" <do-not-reply@qatek04.qatek.corp.dataupia.com>
Date: Tue,  7 Apr 2009 20:36:19 +0000
To: phil@dataupia.com
Subject: System event on qatek04: Paging activity alarm clearing
Return-Path: do-not-reply@qatek04.qatek.corp.dataupia.com
Message-ID: <DTMAIL01NCxLYxrwZnT000028e6@mx2.dataupia.com>
```

Figure F1 Email header from Satori server

These parameters can be configured for all types of emails produced by the blades:

domain	Override domain from which emails appear to come
return-addr	Set the username in the return address for email notifications (default: ‘do-not-reply’)
return-host	Include hostname in return address for email notifications (default: enabled)

There are three possible constructions of the ‘From’ address. These constructions are “evaluated” in the order listed below:

1. “Admin User" <return-addr>. The return path user address, but if it contains a ‘@’ it will be used as the entire email address. Default: ‘do-not-reply’
2. “Admin User" <return-addr@hostname.domain> Recommended. This is the default. This format is selected if ‘return-host is enabled.
3. “Admin User" <return-addr@domain> Not recommended since the return address does not contain the blade’s host name it is difficult to determine which blade was the source of the email. This format is selected if return-host is disabled by issuing the ‘no email return-host’ CLI command.

Hostname is configured by the CLI ‘hostname’ command.

Domain name is configured by the CLI ‘ip domain-list command’ but can be overridden for emails by the CLI ‘email domain’ command.

Configuring administrator emails

These parameters are directly configurable for administrator emails using the email CLI command:

`email notify events`	Specify which events to send notification emails for
`email notify recipients`	Specify which email addresses to send emails to
`email mailhub`	Set the mail relay to be used to send emails
`email mailhub-port`	Set mail port to be used to send emails

Errors encountered in sending emails are logged only to the blade's messages log.

Setting no recipients is equivalent to disabling administrator emails.

Here are some frequently used overrides for administrator emails:

```
no stats alarm cpu_util_indiv enable
stats alarm paging rising clear-threshold 80000
stats alarm paging rising error-threshold 70000
```

The CLI command 'email send-test' will send a test email for the administrator emails, there is no CLI command that will send a test auto-support emails

Configuring auto-support emails

These parameters are configurable for auto-support emails using the email CLI command:

`email auto-support enable`	Send automatic support notifications via email
`email auto-support mailhub`	Set the mail relay to be used to send emails
`email auto-support recipient`	Specify which email addresses to send emails to

Note that the mail hub and recipient list for auto-support emails are entirely unrelated to administrator emails settings. In all other respects, the email headers are constructed in the same manner.

Auto-support emails may contain attachments which exceed a few megabytes in size.

delete

The CLI command 'email send-test' will send a test email for the administrator emails, there is no CLI command that will send a test auto-support emails

Appendix H: Configuring NTP for Satori Server

If the Satori Server blades connect directly to the enterprise time source, usually it is a simple matter to configure one of the Satori client hosts (either a Loader Server or a host Database server) to provide time service to Satori. In order to act as a synchronized time source for Satori Server, the server must have access to the enterprise time service on one interface, and it must have access to Satori Server blades on its second interface.

If there is a Satori Loader server in the configuration, it is usually the best choice to provide time service to Satori Server blades. If this is not an option, most likely one of the host database servers can be used for this purpose (and it probably runs NTP already).

For Satori Loader and DA servers, NTP should be configured on each so that the machine maintains accurate time.

Linux (Loader Server):

Edit /etc/ntp.conf

```
# Permit time synchronization with our time source, but do not
# permit the source to query or modify the service on this system.
restrict default kod nomodify notrap nopeer noquery
restrict -6 default kod nomodify notrap nopeer noquery

# Permit all access over the loopback interface.  This could
# be tightened as well, but to do so would effect some of
# the administrative functions.
restrict 127.0.0.1
restrict -6 ::1

# Hosts on local network are less restricted.
restrict <eth2-interface ip> mask <eth2-interface mask> nomodify notrap

# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (http://www.pool.ntp.org/join.html).
server <ip of ntp time source> dynamic

#broadcast 192.168.1.255 key 42        # broadcast server
#broadcastclient                       # broadcast client
#broadcast 224.0.1.1 key 42            # multicast server
#multicastclient 224.0.1.1             # multicast client
#manycastserver 239.255.254.254        # manycast server
#manycastclient 239.255.254.254 key 42  # manycast client

# Undisciplined Local Clock. This is a fake driver intended for backup
# and when no outside source of synchronized time is available.
server  127.127.1.0     # local clock
fudge   127.127.1.0 stratum 10

# Drift file.  Put this in a directory which the daemon can write to.
# No symbolic links allowed, either, since the daemon updates the file
# by creating a temporary in the same directory and then rename()'ing
# it to the file.
driftfile /var/lib/ntp/drift

# Key file containing the keys and key identifiers used when operating
# with symmetric key cryptography.
keys /etc/ntp/keys
```

```
# Specify the key identifiers which are trusted.
#trustedkey 4 8 42

# Specify the key identifier to use with the ntpdc utility.
#requestkey 8

# Specify the key identifier to use with the ntpq utility.
#controlkey 8
```

Enable the service

```
chkconfig (--levels 345?) ntpd on
service ntpd start
```

Solaris (Host database server)

Request the enterprise database administrator to perform this procedure. This will require root access to the host.

/etc/inet/ntp.conf

```
restrict default nomodify notrap noquery
restrict 127.0.0.1
restrict <Satori-Net-IP> mask <Satori-Net-IP> nomodify notrap
server  127.127.1.0     # local clock
fudge   127.127.1.0 stratum 10
driftfile /var/lib/ntp/ntp.drift
```

Make sure /var/lib/ntp exists

```
mkdir –p /var/lib/ntp
```

Start the service

```
svcadm enable network/ntp
```

Satori Server (blades)

Cli commands:

```
ntp server <IP of time source on DATA net>
ntp enable
```

Appendix I: Satori Server Network Worksheet

Customer Network:

Network address	
Network mask	
Default gateway	
DNS Servers	
Domain Name	
NTP Server	

Satori Network Specs:

Network address	
Network mask	
Default gateway	
DNS Servers	Not used by Satori Server
Domain Name	
NTP Server	

IP Address assignment table (One table should be used per rack)

Hostname/Component	Satori Server Network IP Address	Customer Network IP Address
Host DBMS Servers	**-**	**-**
Loader Servers	**-**	**-**
DA Servers	**-**	**-**
Satori Servers	**-**	**-**
Blade 0/		
Blade 1/		N/C
Blade 2/		N/C
Blade 3/		N/C
Blade 4/		N/C
Blade 5/		N/C
Blade 6/		N/C
Blade 7/		N/C
Blade 8		N/C
Blade 9		N/C
Blade 10		N/C
Blade 11		N/C
Blade 12		N/C
Blade 13		N/C
Blade 14		N/C
Blade 15		N/C
Blade 16		N/C
Peripherals	**-**	**-**
DMC mgmt IP	N/C	
Satori Server mgmt IP		N/A
Satori Switch mgmt IP	N/C	
Term server IP	N/C	
PDU-1 mgmt IP	N/C	

PDU-2 mgmt IP	N/C	

Notes for the Satori Server Network worksheet:

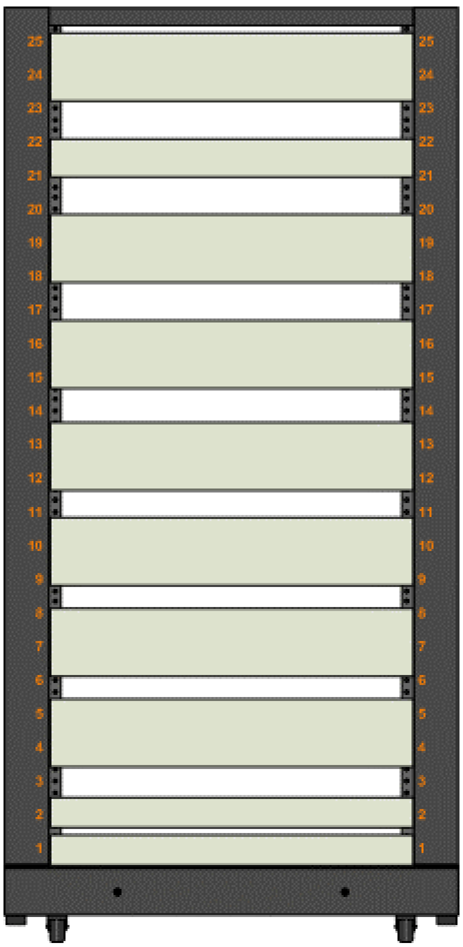
1. Obtain the customer network specs from the customer.
2. Work with the customer's network manager to define the Satori Server network parameters. Refer to the section Satori Server Networking 101. A sample IP configuration can be found here: Sample IP configuration.
3. List the Host DBMS servers, loader servers, DA servers, which will inter-operate with the Satori Server array.
4. For all servers to be installed an IP address on either the customer network (may be an existing IP) and/or Satori Server network must be assigned.
5. All IP's must be static.
6. "Satori Server Management IP" is only used by Satori Server management. Assign any unused address on the Satori Server network.
7. Blade 0 Satori Server is a special case. It has one interface (eth0) attached to the customer network for DMC provisioning. This blade also must have a default route to the customer net default gateway if DMC users are on routed subnets.
8. There is no DNS on the Satori Server network.
9. The Satori Server HEARTBEAT network will self configure (uses zero-conf IP addresses).

Appendix J: Rack layout worksheets

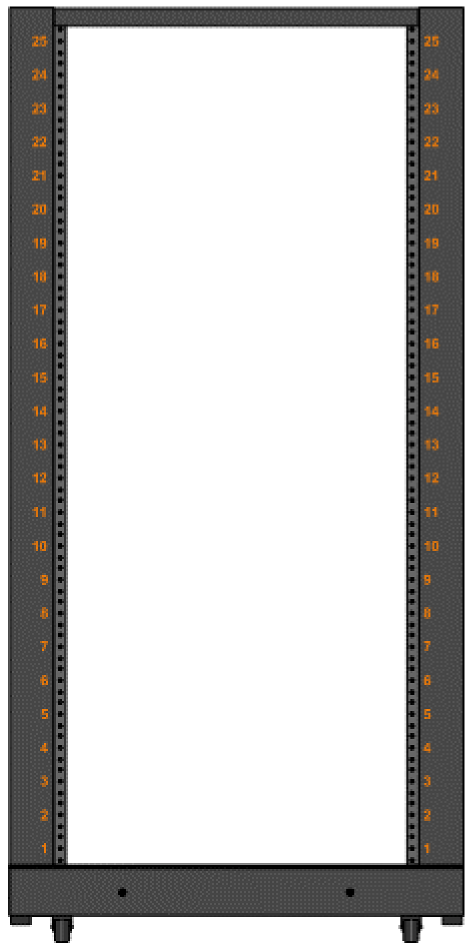
24U APC Rack System

A Dataupia 24U rack system ships with all of the mounting hardware and cabling already installed in the rack. All of the blades are removed and packed separately for shipping, but the network switch, terminal server, and KVM remain in the rack when the system is shipped. The network switch is usually installed at the top of the rack, with the terminal server mounted just below it, and the KVM is usually installed in the middle of the rack. The Power Distribution Units (PDUs) are also shipped installed at the bottom of the rack.

24U Rack



25U Rack

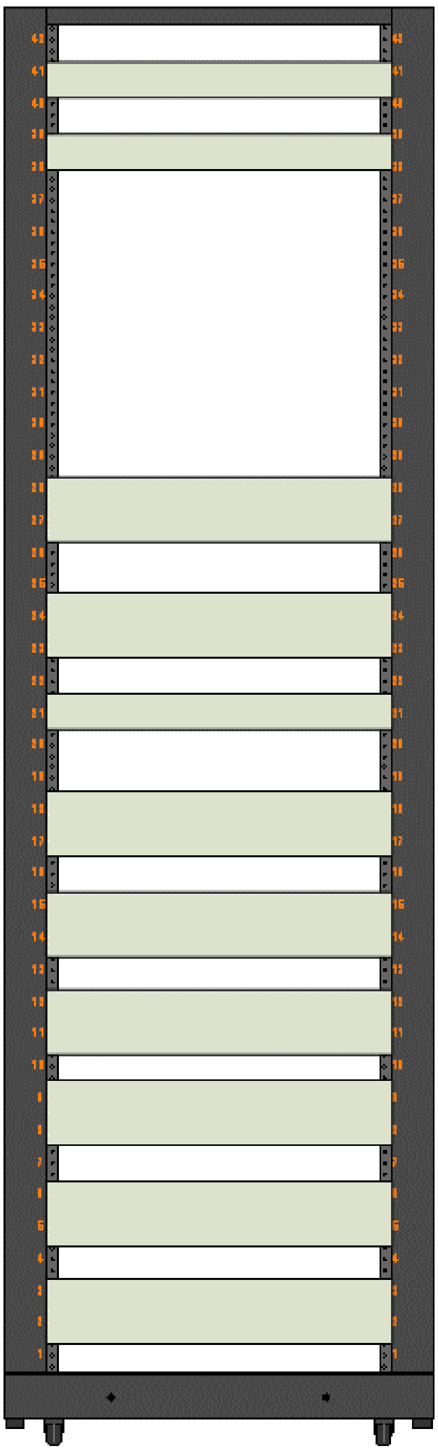


25U Rack

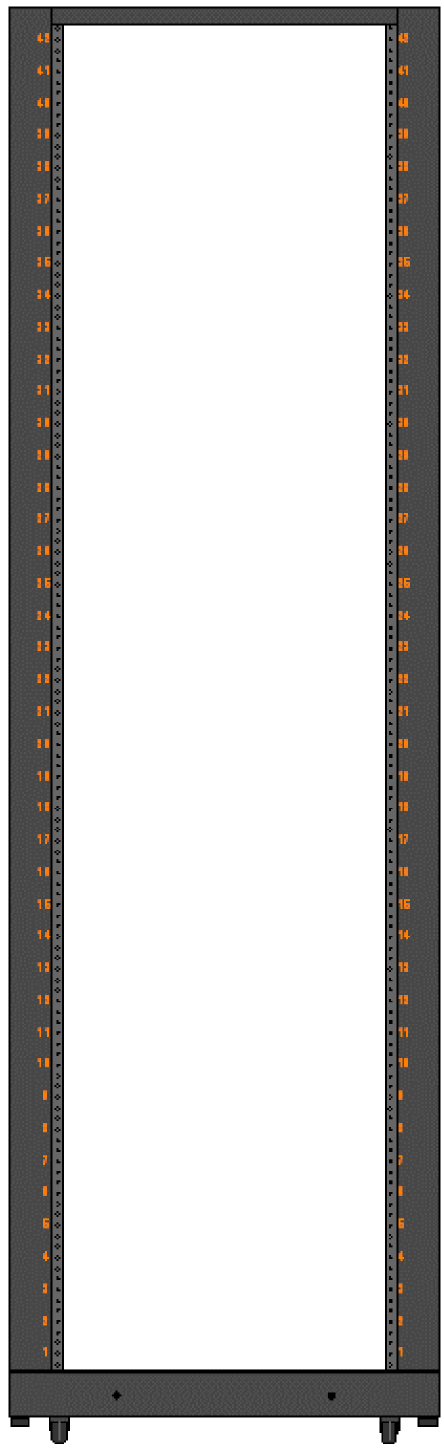
42U APC Rack System

A Dataupia 42U rack system ships with all of the mounting hardware and cabling already installed in the rack. All of the blades are removed and packed separately for shipping, but the network switch, terminal server, and KVM remain in the rack when the system is shipped. The network switch is usually installed at the top of the rack, with the terminal server mounted just below it, and the KVM is usually installed in the middle of the rack. Two vertical Power Distribution Units (PDUs) are also shipped installed at the back of the rack.

42U Rack



42U Rack



42U Rack

Appendix K: Sample Wire Run List

Cable colors and lengths can vary from system to system depending on customer requirements. This information does not constitute a specification, but is provided as an example.

SAMPLE WIRE RUN LIST and SERIAL NUMBERS				
AWG	**Color**	**FROM Component A**	**Length**	**TO Component B**
CAT6'	BLUE	DELL SWITCH PORT 7	3	DA ETH 1
CAT6'	WHITE	DELL SWITCH PORT 23	3	DA ETH 2
CAT6'	GRAY	TERM SERVER PORT 4	3	DA SERIAL
CAT6'	BLUE	DELL SWITCH PORT 3	3	LOADER ETH 1
CAT6'	WHITE	DELL SWITCH PORT 21	3	LOADER ETH 2
CAT6'	GRAY	TERM SERVER PORT 3	3	LOADER SERIAL
CAT6'	WHITE	DELL SWITCH PORT 19	3	SATORI 1 ETH 0
CAT6'	BLUE	DELL SWITCH PORT 1	3	SATORI 1 ETH 1
CAT6'	BLACK	DELL SWITCH PORT 2	3	SATORI 1 ETH 2
CAT6'	GRAY	TERM SERVER PORT 1	3	SATORI 1 SERIAL
CAT6'	BLUE	DELL SWITCH PORT 5	3	SATORI 2 ETH 1
CAT6'	BLACK	DELL SWITCH PORT 4	3	SATORI 2 ETH 2
CAT6'	GRAY	TERM SERVER PORT 2	3	SATORI 2 SERIAL
CAT6'	WHITE	DELL SWITCH PORT 20	5	PDU 1 ETH
CAT6'	WHITE	DELL SWITCH PORT 22	1	TERM ETH
	RED	DELL SWITCH PORT 24	15	CLIENT NIC 1
	GREEN	DELL SWITCH PORT 9	15	CLIENT NIC 2
PWR	BLACK	SATORI 1 HB	6	PDU 1 PORT 5
PWR	BLACK	SATORI 2	6	PDU 1 PORT 6
PWR	BLACK	LOADER SERVER	6	PDU 1 PORT 7
PWR	BLACK	DA SERVER	6	PDU 1 PORT 8
PWR	BLACK	Digicon Term Server	6	PDU 1 PORT 1
PWR	BLACK	Dell 5448 Switch	6	PDU 1 PORT 2
SERIAL NUMBERS				
PDU #1		NA		
SATORI BLADE 2		B0808793		
SATORI BLADE 1HB		641235-08		
DA SERVER		645835-04		
LOADER SERVER		2004554		
TERM SERVER 8 port		SW83614123		
DELL 5448 SWITCH 24 Port		CZM7FK1	Service Tag	
SPARE CHASSIS		B1007123		